# Masked observations

## 1 Defining equations

**Lemma 1.1.** *For every natural number $Q$ and natural number $t$,* $\mathrm{centergate}(Q, t) = \mathrm{not}(\mathrm{center}(t))\,\mathrm{and}\,\mathrm{rightdefect}(Q, t, 1)$.

*Proof.* The goal follows from $\mathrm{centergatedefinition}(Q, t)$. □

**Lemma 1.2.** *For every natural number $Q$ and natural number $t$,* $\mathrm{alignedbackground}(Q, t) = \mathrm{wittshl}(\mathrm{wittrow}(t), Q)$.

*Proof.* The goal follows from $\mathrm{backgrounddefinition}(Q, t)$. □

**Lemma 1.3.** *For every natural number $Q$ and natural number $t$,* $\mathrm{aligneddefect}(Q, t) = \mathrm{wittxor}(\mathrm{wittrow}(t + Q), \mathrm{alignedbackground}(Q, t))$.

*Proof.* The goal follows from $\mathrm{aligneddefectdefinition}(Q, t)$. □

**Lemma 1.4.** *For every natural number $L$ and natural number $H$,* $\mathrm{carryledger}(L, H, 0) = 0$.

*Proof.* The goal follows from $\mathrm{carryledgerdefinition}(L, H)$. □

**Lemma 1.5.** *For every natural number $L$ and natural number $H$ and natural number $t$,* $\mathrm{carryledger}(L, H, \mathrm{succ}(t)) = \mathrm{wittxor}(\mathrm{carryledger}(L, H, t), \mathrm{wittxor}(\mathrm{wittcarry}(\mathrm{page}(L, H, t)), \mathrm{na}$

*Proof.* The goal follows from $\mathrm{carryledgerdefinition2}(L, H, t)$. □

**Lemma 1.6.** *For every natural number $n$,* $\mathrm{centerflipmask}(n) = \mathrm{wittshl}(7, n)$.

*Proof.* The goal follows from $\mathrm{centerflipmaskdefinition}(n)$. □

**Lemma 1.7.** *For every natural number $start$,* $\mathrm{centersign}(start, 0) = 1$.

*Proof.* The goal follows from $\mathrm{centersigndefinition}(start)$. □

**Lemma 1.8.** *For every natural number $start$ and natural number $t$,* $\mathrm{centersign}(start, \mathrm{succ}(t)) = (1 - 2 \cdot \mathrm{bitvalue}(\mathrm{center}(start))) \cdot \mathrm{centersign}(start + 1, t)$.

*Proof.* The goal follows from $\mathrm{centersigndefinition2}(start, t)$. □

**Lemma 1.9.** *For every natural number $b$ and natural number $d$,* $\mathrm{defectstep}(b, d) = \mathrm{wittxor}(\mathrm{wittphi}(d), \mathrm{wittshl}(\mathrm{crosscarry}(b, d), 1))$.

*Proof.* The goal follows from $\mathrm{defectstepdefinition}(b, d)$. □

**Lemma 1.10.** *For every natural number $Q$ and natural number $d$ and natural number $t$,* diagonalphase$(Q, d, t)$ = bitvalue(ldiag$(d, t{+}Q)$)$-$bitvalue(ldiag$(d, t)$).

*Proof.* The goal follows from diagonalphasedefinition$(Q, d, t)$. □

**Lemma 1.11.** *For every natural number $Q$ and natural number $t$,* observersourceflux$(Q, t, 0)$ = 0.

*Proof.* The goal follows from observersourcefluxdefinition$(Q, t)$. □

**Lemma 1.12.** *For every natural number $Q$ and natural number $t$ and natural number $t1$,* observersourceflux$(Q, t, \text{succ}(t1))$ = wittxor(wittlinear(observersourceflux$(Q, t, t1)$), wittshl($t1), t1 + 1)$).

*Proof.* The goal follows from observersourcefluxdefinition2$(Q, t, t1)$. □

**Lemma 1.13.** *For every natural number $L$ and natural number $H$,* pageboundarycarry$(L, H, 0)$ = false.

*Proof.* The goal follows from pageboundarycarrydefinition$(L, H)$. □

**Lemma 1.14.** *For every natural number $L$ and natural number $H$ and natural number $t$,* pageboundarycarry$(L, H, \text{succ}(t))$ = rowbit(page$(L, H, t + 1), 0)$ and rowbit(page$(L, H, t), L - 1)$.

*Proof.* The goal follows from pageboundarycarrydefinition2$(L, H, t)$. □

**Lemma 1.15.** *For every natural number $L$ and natural number $H$,* pagefold$(L, H, 0)$ = 0.

*Proof.* The goal follows from pagefolddefinition$(L, H)$. □

**Lemma 1.16.** *For every natural number $L$ and natural number $H$ and natural number $t$,* pagefold$(L, H, \text{succ}(t))$ = wittxor(pagefold$(L, H, t)$, page$(L, H, t)$).

*Proof.* The goal follows from pagefolddefinition2$(L, H, t)$. □

**Lemma 1.17.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* pathflux$(Q, t, k, 0) = 0$.

*Proof.* The goal follows from pathfluxdefinition$(Q, t, k)$. □

**Lemma 1.18.** *For every natural number $d$,* qordersearch$(d, 0) = 0$.

*Proof.* The goal follows from qordersearchdefinition$(d)$. □

**Lemma 1.19.** *For every natural number $t$ and natural number $k$,* $\mathrm{rayflux}(t, k, 0) = 0$.

*Proof.* The goal follows from $\mathrm{rayfluxdefinition}(t, k)$. □

**Lemma 1.20.** *For every truth value $a$ and truth value $g$ and integer $outgoing$ and integer $gate$,* $\mathrm{recoverphase}(a, g, outgoing, gate) = (1 - 2 \cdot \mathrm{bitvalue}(g)) \cdot (outgoing - (1 - 2 \cdot \mathrm{bitvalue}(a)) \cdot gate)$.

*Proof.* The goal follows from $\mathrm{recoverphasedefinition}(a, g, outgoing, gate)$. □

**Lemma 1.21.** *For every natural number $t$,* $\mathrm{rightimport}(t) = \mathrm{not}(\mathrm{rightcell}(t, 1))\,\mathrm{and}\,\mathrm{rightcell}(t, 2)$.

*Proof.* The goal follows from $\mathrm{rightimportdefinition}(t)$. □

**Lemma 1.22.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* $\mathrm{rightphase}(Q, t, k) = \mathrm{bitvalue}(\mathrm{rightcell}(t{+}Q, k)){-}\mathrm{bitvalue}(\mathrm{rightcell}(t, k))$.

*Proof.* The goal follows from $\mathrm{rightphasedefinition}(Q, t, k)$. □

**Lemma 1.23.** *For every truth value $a$ and truth value $g$ and integer $incoming$ and integer $gate$,* $\mathrm{transferphase}(a, g, incoming, gate) = (1 - 2 \cdot \mathrm{bitvalue}(g)) \cdot incoming + (1 - 2 \cdot \mathrm{bitvalue}(a)) \cdot gate$.

*Proof.* The goal follows from $\mathrm{transferphasedefinition}(a, g, incoming, gate)$. □

**Lemma 1.24.** *For every natural number $n$,* $\mathrm{twopow}(\mathrm{succ}(n)) = \mathrm{twopow}(n){+}\,\mathrm{twopow}(n)$.

*Proof.* The goal follows from $\mathrm{twopowdefinition2}(n)$. □

**Lemma 1.25.** *For every natural number $y$,* $\mathrm{wittlinear}(y) = \mathrm{wittxor}(\mathrm{wittxor}(y, \mathrm{wittshl}(y, 1)), \mathrm{wittshl}(y, 2$

*Proof.* The goal follows from $\mathrm{philineardefinition}(y)$. □

**Lemma 1.26.** *For every natural number $h1$ and natural number $m$,* $\mathrm{wittlinpow}(\mathrm{succ}(m), h1) = \mathrm{wittlinear}(\mathrm{wittlinpow}(m, h1))$.

*Proof.* The goal follows from $\mathrm{philinearpowdefinition2}(h1, m)$. □

**Lemma 1.27.** *For every natural number $n$,* $\mathrm{wittrow}(\mathrm{succ}(n)) = \mathrm{wittphi}(\mathrm{wittrow}(n))$.

*Proof.* The goal follows from $\mathrm{rowdefinition2}(n)$. □

**Lemma 1.28.** *For every natural number $Q$ and natural number $t$ and natural number $i$,* $\mathrm{alignedphase}(Q, t, i) = \mathrm{pairphase}(\mathrm{alignedbackground}(Q, t), \mathrm{aligneddefect}(Q, t), i)$.

*Proof.* The goal follows from alignedphasedefinition$(Q, t, i)$. □

**Lemma 1.29.** *For every natural number $h$ and natural number $A$,* blockendpoint$(h, A) =$ rowbit$(A, 0)$ and rowbit$(A, h - 1)$.

*Proof.* The goal follows from blockendpointdefinition$(h, A)$. □

**Lemma 1.30.** *For every natural number $t$,* center$(t) =$ cell$(t, 0)$.

*Proof.* The goal follows from centerdefinition$(t)$. □

**Lemma 1.31.** *For every natural number $Q$ and natural number $d$ and natural number $t$ and natural number $p$,* diagonalflux$(Q, d, t, p) =$ rayflux$(t + Q, t + Q - d, t + Q - d - p) -$ rayflux$(t, t - d, t - d - p)$.

*Proof.* The goal follows from diagonalfluxdefinition$(Q, d, t, p)$. □

**Lemma 1.32.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* erasureevent$(Q, t, k) =$ (leftdefect$(Q, t, k+1) =$ true $\wedge$ leftdefect$(Q, t+1, k) =$ false $\wedge$ gatecode$(Q, t, k) \neq 0 \wedge$collisioncode$(Q, t, k) = 0 \wedge$(leftcolour$(Q, t+1, k) = 0 \vee$ leftcolour$(Q, t + 1, k) = 3)$).

*Proof.* The goal follows from erasureeventdefinition$(Q, t, k)$. □

**Lemma 1.33.** ldiag$(0, 0) =$ true.

*Proof.* The goal follows from ldiagdefinition. □

**Lemma 1.34.** *For every natural number $n$,* ldiag$(\text{succ}(n), 0) =$ false.

*Proof.* The goal follows from ldiagdefinition2$(n)$. □

**Lemma 1.35.** *For every natural number $s$,* ldiag$(0, \text{succ}(s)) =$ true.

*Proof.* The goal follows from ldiagdefinition3$(s)$. □

**Lemma 1.36.** *For every natural number $s$,* ldiag$(\text{succ}(0), \text{succ}(s)) =$ true.

*Proof.* The goal follows from ldiagdefinition4$(s)$. □

**Lemma 1.37.** *For every natural number $s$ and natural number $n1$,* ldiag$(\text{succ}(\text{succ}(n1)), \text{succ}(s)) =$ bxor(ldiag$(n1, s)$, ldiag$(\text{succ}(n1), s)$ or ldiag$(\text{succ}(\text{succ}(n1)), s)$).

*Proof.* The goal follows from ldiagdefinition5$(s, n1)$. □

**Lemma 1.38.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* $\mathrm{leadingadvance}(Q, t, k) = (2 \leq k \wedge \mathrm{advancingatdepth}(Q, t, k) \wedge \mathrm{rightmostleft}(Q, t, k))$.

*Proof.* The goal follows from $\mathrm{leadingadvancedefinition}(Q, t, k)$. □

**Lemma 1.39.** *For every truth value $b$,* $\mathrm{naturalbit}(b) = \mathrm{naturalbit}(b)$.

*Proof.* The goal follows from $\mathrm{naturalbitdefinition}(b)$. □

**Lemma 1.40.** *For every natural number $t$,* $\mathrm{observerinput}(t) = \mathrm{not}(\mathrm{center}(t))$ and $\mathrm{rightcell}(t, 1)$.

*Proof.* The goal follows from $\mathrm{observerinputdefinition}(t)$. □

**Lemma 1.41.** *For every natural number $Q$ and natural number $t$,* $\mathrm{observerqrank}(Q, t) = t + Q - 2 \cdot \mathrm{qorder}(\mathrm{observerdefect}(Q, t))$.

*Proof.* The goal follows from $\mathrm{observerqrankdefinition}(Q, t)$. □

**Lemma 1.42.** *For every natural number $b$ and natural number $d$ and natural number $i$,* $\mathrm{pairphase}(b, d, i) = \mathrm{bitvalue}(\mathrm{rowbit}(\mathrm{wittxor}(b, d), i)) - \mathrm{bitvalue}(\mathrm{rowbit}(b, i))$.

*Proof.* The goal follows from $\mathrm{pairphasedefinition}(b, d, i)$. □

**Lemma 1.43.** *For every natural number $a$ and natural number $i$,* $\mathrm{rowbit}(a, i) = \mathrm{rowbit}(a, i)$.

*Proof.* The goal follows from $\mathrm{rowbitdefinition}(a, i)$. □

**Lemma 1.44.** *For every natural number $h1$,* $\mathrm{wittwin}(h1, 0) = 0$.

*Proof.* The goal follows from $\mathrm{winwittdefinition}(h1)$. □

**Lemma 1.45.** *For every natural number $h1$ and natural number $s$,* $\mathrm{wittwin}(h1, \mathrm{succ}(s)) = \mathrm{wittxor}(\mathrm{wittlinear}(\mathrm{wittwin}(h1, s)), \mathrm{wittshl}(\mathrm{wittcarry}(\mathrm{wittrow}(h1 + s)), 1))$.

*Proof.* The goal follows from $\mathrm{winwittdefinition2}(h1, s)$. □

**Lemma 1.46.** *For every natural number $n$ and natural number $j$ and truth value $b$,* $\mathrm{bitcarryeq}(n, j, b) = (\mathrm{wittbit}(\mathrm{wittcarry}(n), j) = \mathrm{bton}(b))$.

*Proof.* The goal follows from $\mathrm{bitcarryeqdefinition}(n, j, b)$. □

**Lemma 1.47.** *For every natural number $a$ and natural number $b$ and natural number $j$ and truth value $c$,* $\mathrm{bitcrosscarryeq}(a, b, j, c) = (\mathrm{wittbit}(\mathrm{crosscarry}(a, b), j) = \mathrm{bton}(c))$.

*Proof.* The goal follows from bitcrosscarryeqdefinition$(a, b, j, c)$. □

**Lemma 1.48.** *For every natural number $a$ and natural number $b$ and natural number $j1$ and natural number $j2$ and truth value $c$,* bitcrosscarryxoreq$(a, b, j1, j2, c) =$ (bxor(rowbit(crosscarry$(a, b), j1)$, rowbit(crosscarry$(a, b), j2)) = c$).

*Proof.* The goal follows from bitcrosscarryxoreqdefinition$(a, b, j1, j2, c)$. □

**Lemma 1.49.** *For every natural number $k$ and natural number $y$ and natural number $j$ and truth value $b$,* bitfrobdefecteq$(k, y, j, b) =$ (wittbit(frobdefect$(k, y), j) =$ bton$(b)$).

*Proof.* The goal follows from bitfrobdefecteqdefinition$(k, y, j, b)$. □

**Lemma 1.50.** *For every natural number $t0$ and natural number $m$ and natural number $j$ and truth value $b$,* bitwinwitteq$(t0, m, j, b) =$ (wittbit(wittwin$(t0, m), j) =$ bton$(b)$).

*Proof.* The goal follows from bitwinwitteqdefinition$(t0, m, j, b)$. □

**Lemma 1.51.** *For every natural number $k$ and natural number $y$,* frobdefect$(k, y) =$ wittxor(wittcarry(wittscale$(k, y))$, wittscale$(k,$ wittcarry$(y)))$.

*Proof.* The goal follows from frobdefectdefinition$(k, y)$. □

**Lemma 1.52.** *For every natural number $t0$,* maskcount$(t0, 0) =$ false.

*Proof.* The goal follows from maskcountdefinition$(t0)$. □

**Lemma 1.53.** *For every natural number $t$ and natural number $k$,* maskparity$(t, k) =$ maskcount$(t,$ twopow$(k))$.

*Proof.* The goal follows from maskparitydefinition$(t, k)$. □

**Lemma 1.54.** *For every natural number $t$,* rcenter$(t) =$ rdiag$(t, t)$.

*Proof.* The goal follows from rcenterdefinition$(t)$. □

**Lemma 1.55.** rdiag$(0, 0) =$ true.

*Proof.* The goal follows from rdiagdefinition. □

**Lemma 1.56.** *For every natural number $n$,* rdiag(succ$(n), 0) =$ false.

*Proof.* The goal follows from rdiagdefinition2$(n)$. □

**Lemma 1.57.** *For every natural number $s$,* rdiag$(0,$ succ$(s)) =$ true.

*Proof.* The goal follows from rdiagdefinition3($s$). □

**Lemma 1.58.** *For every natural number $s$,* rdiag(succ(0), succ($s$)) = not(rdiag(succ(0), $s$)).

*Proof.* The goal follows from rdiagdefinition4($s$). □

**Lemma 1.59.** *For every natural number $s$ and natural number $n1$,* rdiag(succ(succ($n1$)), succ($s$)) = bxor(rdiag(succ(succ($n1$)), $s$), rdiag(succ($n1$), $s$) or rdiag($n1$, $s$)).

*Proof.* The goal follows from rdiagdefinition5($s$, $n1$). □

**Lemma 1.60.** *For every natural number $t$ and natural number $k$,* response($t$, $k$) = window($t$, twopow($k$), 0).

*Proof.* The goal follows from responsedefinition($t$, $k$). □

**Lemma 1.61.** *For every truth value $a$ and truth value $b$ and truth value $c$,* rule($a$, $b$, $c$) = bxor($a$, $b$ or $c$).

*Proof.* The goal follows from ruledefinition($a$, $b$, $c$). □

**Lemma 1.62.** twopow(0) = 1.

*Proof.* The goal follows from twopowdefinition. □

**Lemma 1.63.** *For every natural number $h1$ and integer $h2$,* window($h1$, 0, $h2$) = false.

*Proof.* The goal follows from window60definition($h1$, $h2$). □

**Lemma 1.64.** *For every natural number $y$ and natural number $j$,* wittbitbool($y$, $j$) = rowbit($y$, $j$).

*Proof.* The goal follows from wittbitbooldefinition($y$, $j$). □

**Lemma 1.65.** *For every natural number $t$,* wittcenter($t$) = wittbit(wittrow($t$), $t$).

*Proof.* The goal follows from rowcenterdefinition($t$). □

**Lemma 1.66.** *For every natural number $h1$,* wittlinpow(0, $h1$) = $h1$.

*Proof.* The goal follows from philinearpowdefinition($h1$). □

**Lemma 1.67.** *For every natural number $t$ and natural number $k$,* wittresp($t$, $k$) = rowbit(wittwin($t$, twopow($k$)), $t$ + twopow($k$)).

*Proof.* The goal follows from wittrespbooldefinition($t$, $k$). □

**Lemma 1.68.** wittrow$(0) = 1$.

*Proof.* The goal follows from rowdefinition. □

**Lemma 1.69.** *For every natural number $k$ and natural number $y$,* wittscale$(k, y) =$ wittxor(wittxor$(y,$ wittshl$(y,$ twopow$(k))),$ wittshl$(y,$ twopow$(k)+$twopow$(k)))$.

*Proof.* The goal follows from phiscaledefinition$(k, y)$. □

**Lemma 1.70.** *For every natural number $t$,* ledgerdiagonal$(t) =$ rowbit(wittwin$(0, t), t)$.

*Proof.* The goal follows from ledgerdiagonaldefinition$(t)$. □

**Lemma 1.71.** *For every natural number $t$ and natural number $j$,* ledgerbit$(t, j) =$ rowbit(wittwin$(0, t), j)$.

*Proof.* The goal follows from ledgerbitdefinition$(t, j)$. □

**Lemma 1.72.** *For every natural number $t$,* edgediagonal$(t) =$ rowbit(edgeledger$(t), t)$.

*Proof.* The goal follows from edgediagonaldefinition$(t)$. □

**Lemma 1.73.** *For every natural number $t$,* interiordiagonal$(t) =$ rowbit(interiorledger$(t), t)$.

*Proof.* The goal follows from interiordiagonaldefinition$(t)$. □

**Lemma 1.74.** *For every natural number $t$,* interior$(t) =$ wittxor(wittcarry(wittrow$(t))$, edgesource$(t))$.

*Proof.* The goal follows from interiordefinition$(t)$. □

## 2 Masked observations

**Theorem 2.1.** *For every natural number $t$,* whitecount$(t, 0) = 0$.

*Proof.* The goal follows from whitecountzero$(t)$. □

**Theorem 2.2.** *For every natural number $t$ and natural number $n$, if* center$(t) =$ true*, then* whitecount$(t, n + 1) =$ whitecount$(t + 1, n)$.

*Proof.* The goal follows from whitecountblackstep$(t, n)$. □

**Theorem 2.3.** *For every natural number $t$ and natural number $n$, if* center$(t) =$ false*, then* whitecount$(t, n + 1) =$ whitecount$(t + 1, n) + 1$.

*Proof.* The goal follows from whitecountwhitestep$(t, n)$. □

**Theorem 2.4.** *For every natural number $t1$ and natural number $t2$ and natural number $n$, if* centerwindowagree$(t1, t2, n)$*, then* leftprefixagree$(t1, t2, n)$ *if and only if* whitetraceagree$(t1, t2, n)$.

*Proof.* The goal follows from prefixagreementiffwhitetrace$(t1, t2, n)$. □

**Theorem 2.5.** *For every natural number $T$ and natural number $P$ and natural number $t$ and natural number $n$, if $0 < P$, then if* centerperiodic$(T, P)$*, then if $T \leq t$, then there exists a natural number $a$ such that there exists a natural number $b$ such that $a < b$ and $b \leq$* twopow(whitecount$(t, n)) \wedge$ leftprefixagree$(t + a \cdot P, t + b \cdot P, n)$.

*Proof.* The goal follows from compressedprefixrepeat$(T, P, t, n)$. □

**Theorem 2.6.** *For every natural number $T$ and natural number $P$ and natural number $t$ and natural number $n$ and natural number $M$, if $0 < P$, then if* centerperiodic$(T, P)$*, then if $T \leq t$, then if* prefixseparated$(t, P, n, M)$*, then $M \leq$* twopow(whitecount$(t, n)$).

*Proof.* The goal follows from separatedprefixbound$(T, P, t, n, M)$. □

**Theorem 2.7.** *For every natural number $T$ and natural number $P$ and natural number $t$ and natural number $n$, if $0 < P$, then if* centerperiodic$(T, P)$*, then if $T \leq t$, then $n < t +$* twopow(whitecount$(t, n)) \cdot P$.

*Proof.* The goal follows from edgeinformationbound$(T, P, t, n)$. □

**Theorem 2.8.** *For every natural number $n$ and natural number $t$ and natural number $w$ and natural number $P$, if $n < (w+1) \cdot P$, then $n < t +$*twopow$(w) \cdot P$.

*Proof.* The goal follows from informationboundaudit$(n, t, w, P)$. □

## 3 Central collision recurrence

**Theorem 3.1.** *For every natural number $Q$ and natural number $t$ and natural number $r$, if* leftadjacentdefect$(Q, t) =$ true*, then if* center$(t + r) =$ true*, then if* center$(t + r + Q) =$ true*, then* nearcollision$(Q, t, r + 1, 2 \cdot (r + 1))$.

*Proof.* The goal follows from unitdefectforcesnearcollision$(Q, t, r)$. □

**Theorem 3.2.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $T \leq t$, then if* leftadjacentdefect$(m \cdot P, t) =$ true*, then* nearcollision$(m \cdot P, t, P, 2 \cdot P)$.

*Proof.* The goal follows from periodicunitdefectforcesnearcollision$(T, P, m, t)$. □

**Theorem 3.3.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $T \leq N$, then there exists a natural number $t$ such that $N \leq t$ and $t+1 \leq 2 \cdot N + m \cdot P + P \wedge$* centralcollision$(m \cdot P, t, 2 \cdot P)$.

*Proof.* The goal follows from periodiccentralcollisionwindow$(T, P, m, N)$. □

**Theorem 3.4.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then there exists a natural number $t$ such that $N \leq t$ and* centralcollision$(m \cdot P, t, 2 \cdot P)$.

*Proof.* The goal follows from periodiccentralcollisions$(T, P, m, N)$. □

**Theorem 3.5.** *For every natural number $T$ and natural number $P$ and natural number $m$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then* recurrentcentralcollision$(m \cdot P, 2 \cdot P)$.

*Proof.* The goal follows from periodicrecurrentcentralcolumn$(T, P, m)$. □

**Theorem 3.6.** *For every natural number $Q$ and natural number $b$ and natural number $r$, if $0 < r$, then if* centerwindowagree$(b, b + Q, r + 1)$*, then if* center$(b) =$ true*, then if* leftadjacentdefect$(Q, b+r) =$ true*, then there exists a natural number $s$ such that $2 \cdot s \leq r+1$ and $s \leq r \wedge$*leftcollision$(Q, b+s, r+1)$.

*Proof.* The goal follows from blackwindowforcesleftcollision$(Q, b, r)$. □

**Theorem 3.7.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $T \leq N$, then there exists a natural number $t$ such that $N \leq t$ and $t + 1 \leq 2 \cdot N + m \cdot P + 2 \cdot P \wedge$* leftcollision$(m \cdot P, t, P)$.

*Proof.* The goal follows from periodicleftcollisionwindow$(T, P, m, N)$. □

**Theorem 3.8.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then there exists a natural number $t$ such that $N \leq t$ and* leftcollision$(m \cdot P, t, P)$.

*Proof.* The goal follows from periodicleftcollisions$(T, P, m, N)$. □

**Theorem 3.9.** *For every natural number $T$ and natural number $P$ and natural number $m$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then* recurrentleftcollision$(m \cdot P, P)$.

*Proof.* The goal follows from periodicrecurrentleftcolumn$(T, P, m)$. □

**Theorem 3.10.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* collisioncode$(Q, t, k) \neq 0$ *if and only if* collisionbit$(Q, t, k) =$ true.

*Proof.* The goal follows from collisioncodenonzero$(Q, t, k)$. □

**Theorem 3.11.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* collisioncode$(Q, t, k) \leq 6$.

*Proof.* The goal follows from collisioncodebound$(Q, t, k)$. □

**Theorem 3.12.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $0 < k$, then* collisioncode$(Q, t, k) = 1 \leftrightarrow$ leftcolour$(Q, t, k) =$ $1 \wedge$leftcolour$(Q, t, k-1) = 2$ *and* (collisioncode$(Q, t, k) = 2 \leftrightarrow$ leftcolour$(Q, t, k) =$ $2 \wedge$leftcolour$(Q, t, k-1) = 1) \wedge$(collisioncode$(Q, t, k) = 3 \leftrightarrow$ leftcolour$(Q, t, k) =$ $1 \wedge$leftcolour$(Q, t, k-1) = 3) \wedge$(collisioncode$(Q, t, k) = 4 \leftrightarrow$ leftcolour$(Q, t, k) =$ $3 \wedge$leftcolour$(Q, t, k-1) = 1) \wedge$(collisioncode$(Q, t, k) = 5 \leftrightarrow$ leftcolour$(Q, t, k) =$ $2 \wedge$leftcolour$(Q, t, k-1) = 3) \wedge$(collisioncode$(Q, t, k) = 6 \leftrightarrow$ leftcolour$(Q, t, k) =$ $3 \wedge$ leftcolour$(Q, t, k-1) = 2)$.

*Proof.* The goal follows from collisioncodecolours$(Q, t, k)$. □

**Theorem 3.13.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* advancingatdepth$(Q, t, k)$ *if and only if* collisioncode$(Q, t, k) =$ $3$ *or* collisioncode$(Q, t, k) = 5$.

*Proof.* The goal follows from advancingcollisioncodes$(Q, t, k)$. □

**Theorem 3.14.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* collisioncode$(Q, t, k) \neq 0$*, then* leftdefect$(Q, t+1, k) =$ leftdefect$(Q, t, k+1)$.

*Proof.* The goal follows from collisiontransmitsleft$(Q, t, k)$. □

**Theorem 3.15.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* collisioncode$(Q, t, k) \neq 0$*, then* leftgatedifference$(Q, t, k) =$ false.

*Proof.* The goal follows from collisiongatesagree$(Q, t, k)$. □

**Theorem 3.16.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $2 \leq k$, *then* $\text{collisioncode}(Q, t, k) = 1 \to \text{leftdefect}(Q, t+1, k-1) = \text{leftcell}(t, k-2)$ *and* $(\text{collisioncode}(Q, t, k) = 2 \to \text{leftdefect}(Q, t+1, k-1) = \text{leftcell}(t+Q, k-2)) \wedge (\text{collisioncode}(Q, t, k) = 3 \to \text{leftdefect}(Q, t+1, k-1) = \text{true}) \wedge (\text{collisioncode}(Q, t, k) = 4 \to \text{leftdefect}(Q, t+1, k-1) = \text{not}(\text{leftcell}(t+Q, k-2))) \wedge (\text{collisioncode}(Q, t, k) = 5 \to \text{leftdefect}(Q, t+1, k-1) = \text{true}) \wedge (\text{collisioncode}(Q, t, k) = 6 \to \text{leftdefect}(Q, t+1, k-1) = \text{not}(\text{leftcell}(t, k-2)))$.

*Proof.* The goal follows from $\text{collisionrighttransporttable}(Q, t, k)$. □

**Theorem 3.17.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $0 < k$, *then* $\text{collisioncode}(Q, t, k) = 3 \to \text{leftcolour}(Q, t+1, k-1) = 2$ *and if* $\text{collisioncode}(Q, t, k) = 5$, *then* $\text{leftcolour}(Q, t+1, k-1) = 1$.

*Proof.* The goal follows from $\text{advancingrightcolourtable}(Q, t, k)$. □

**Theorem 3.18.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* $\text{leftdefect}(Q, t+1, k) = \text{not}(\text{leftdefect}(Q, t, k+1))$ *if and only if* $\text{leftgatedifference}(Q, t, k) = \text{true}$.

*Proof.* The goal follows from $\text{gatedifferenceiffflip}(Q, t, k)$. □

**Theorem 3.19.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $\text{leftdefect}(Q, t, k+1) = \text{true}$, *then* $\text{leftdefect}(Q, t+1, k) = \text{false}$ *if and only if* $\text{leftgatedifference}(Q, t, k) = \text{true}$.

*Proof.* The goal follows from $\text{incomingdefecterasediff}(Q, t, k)$. □

**Theorem 3.20.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $\text{leftdefect}(Q, t, k+1) = \text{true}$, *then if* $\text{leftdefect}(Q, t+1, k) = \text{false}$, *then* $\text{collisioncode}(Q, t, k) = 0$.

*Proof.* The goal follows from $\text{erasurehasnocollision}(Q, t, k)$. □

**Theorem 3.21.** *For every natural number $Q$ and natural number $k$, if* $0 < k$, *then* $\text{leftcolour}(Q, 0, k) = 0$ *or* $\text{leftcolour}(Q, 0, k) = 2$.

*Proof.* The goal follows from $\text{seedleftcolours}(Q, k)$. □

**Theorem 3.22.** *For every natural number $Q$ and natural number $k$, if* $2 \leq k$, *then* $\text{collisioncode}(Q, 0, k) = 0$.

*Proof.* The goal follows from $\text{seedleftcollisionzero}(Q, k)$. □

**Theorem 3.23.** collisioncode$(1, 1, 2) = 5$.

*Proof.* The goal follows from seedadvancingcollisionbirth. □

**Theorem 3.24.** leftdefect$(3, 0, 3) = \text{true}$ *and* leftdefect$(3, 1, 2) = \text{false} \wedge$ leftgatedifference$(3, 0, 2) = \text{true}$.

*Proof.* The goal follows from seedgateerasure. □

**Theorem 3.25.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* leftdefect$(Q, t, k) = \text{true}$*, then if* center$(t) = \text{true}$*, then if* center$(t + Q) = \text{true}$*, then* advancingleftcone$(Q, t, k)$.

*Proof.* The goal follows from leftdefectblackcone$(Q, t, k)$. □

**Theorem 3.26.** *For every natural number $Q$ and natural number $b$ and natural number $r$, if $0 < r$, then if* centerwindowagree$(b, b + Q, r + 1)$*, then if* center$(b) = \text{true}$*, then if* leftadjacentdefect$(Q, b+r) = \text{true}$*, then there exists a natural number $s$ such that $2 \cdot s \leq r+1$ and $s \leq r \wedge$*advancingleftcollision$(Q, b+s, r + 1)$.

*Proof.* The goal follows from blackwindowforcesadvancingcollision$(Q, b, r)$. □

**Theorem 3.27.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $T \leq N$, then there exists a natural number $t$ such that $N \leq t$ and $t + 1 \leq 2 \cdot N + m \cdot P + 2 \cdot P \wedge$* advancingleftcollision$(m \cdot P, t, P)$.

*Proof.* The goal follows from periodicadvancingleftcollisionwindow$(T, P, m, N)$. □

**Theorem 3.28.** *For every natural number $T$ and natural number $P$ and natural number $m$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then* recurrentadvancingleftcollision$(m \cdot P, P)$.

*Proof.* The goal follows from periodicrecurrentadvancingleftcolumn$(T, P, m)$. □

**Theorem 3.29.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* leftdefect$(Q, t, k) = \text{false}$ *if and only if* leftcolour$(Q, t, k) = 0$ *or* leftcolour$(Q, t, k) = 3$.

*Proof.* The goal follows from leftdefectfalsecolours$(Q, t, k)$. □

**Theorem 3.30.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* $\mathrm{leftdefect}(Q, t, k) = \mathrm{true}$ *if and only if* $\mathrm{leftcolour}(Q, t, k) = 1$ *or* $\mathrm{leftcolour}(Q, t, k) = 2$.

*Proof.* The goal follows from $\mathrm{leftdefecttruecolours}(Q, t, k)$. □

**Theorem 3.31.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $t < k$, then* $\mathrm{leftcolour}(Q, t, k) = 0$ *or* $\mathrm{leftcolour}(Q, t, k) = 2$.

*Proof.* The goal follows from $\mathrm{leftcoloursoutsideseedcone}(Q, t, k)$. □

**Theorem 3.32.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $\mathrm{collisioncode}(Q, t, k) \neq 0$, *then* $k \leq t + 1$.

*Proof.* The goal follows from $\mathrm{collisioncodesupport}(Q, t, k)$. □

**Theorem 3.33.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* $\mathrm{gatecode}(Q, t, k) \neq 0$ *if and only if* $\mathrm{leftgatedifference}(Q, t, k) = \mathrm{true}$.

*Proof.* The goal follows from $\mathrm{gatecodenonzero}(Q, t, k)$. □

**Theorem 3.34.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* $\mathrm{gatecode}(Q, t, k) \leq 6$.

*Proof.* The goal follows from $\mathrm{gatecodebound}(Q, t, k)$. □

**Theorem 3.35.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $0 < k$, then* $\mathrm{gatecode}(Q, t, k) = 1 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 1 \wedge \mathrm{leftcolour}(Q, t, k - 1) = 0$ *and* $(\mathrm{gatecode}(Q, t, k) = 2 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 0 \wedge \mathrm{leftcolour}(Q, t, k-1) = 1) \wedge (\mathrm{gatecode}(Q, t, k) = 3 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 1 \wedge \mathrm{leftcolour}(Q, t, k-1) = 1) \wedge (\mathrm{gatecode}(Q, t, k) = 4 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 2 \wedge \mathrm{leftcolour}(Q, t, k-1) = 0) \wedge (\mathrm{gatecode}(Q, t, k) = 5 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 0 \wedge \mathrm{leftcolour}(Q, t, k-1) = 2) \wedge (\mathrm{gatecode}(Q, t, k) = 6 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 2 \wedge \mathrm{leftcolour}(Q, t, k - 1) = 2)$.

*Proof.* The goal follows from $\mathrm{gatecodecolours}(Q, t, k)$. □

**Theorem 3.36.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $\mathrm{leftdefect}(Q, t, k + 1) = \mathrm{true}$, *then if* $\mathrm{leftdefect}(Q, t + 1, k) = \mathrm{false}$, *then* $\mathrm{erasureevent}(Q, t, k)$.

*Proof.* The goal follows from $\mathrm{erasureeventofstep}(Q, t, k)$. □

**Theorem 3.37.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* erasureevent$(Q, t, k)$, *then* leftcolour$(Q, t + 1, k) = 3$ *if and only if* gatecode$(Q, t, k) \leq 3$ *and* leftcolour$(Q, t, k + 1) = 2$ *or* $4 \leq$ gatecode$(Q, t, k)$ *and* leftcolour$(Q, t, k + 1) = 1$.

*Proof.* The goal follows from erasureblackphaseiff$(Q, t, k)$. □

**Theorem 3.38.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $t+1 < k$, then if* erasureevent$(Q, t, k)$, *then* leftcolour$(Q, t+1, k) = 0$.

*Proof.* The goal follows from erasureoutsideseedconewhite$(Q, t, k)$. □

**Theorem 3.39.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $2 \leq k$, then if* advancingatdepth$(Q, t, k)$, *then if* leftdefect$(Q, t+k, 0) =$ false, *then there exists a natural number $j$ such that* firstpatherasure$(Q, t, k, j)$.

*Proof.* The goal follows from advancingsourcefirsterasure$(Q, t, k)$. □

**Theorem 3.40.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$, *then if $T \leq N$, then there exists a natural number $u$ such that there exists a natural number $d$ such that $N \leq u$ and $u + 1 \leq 2 \cdot N + m \cdot P + 3 \cdot P \wedge d + 2 \leq P \wedge$* erasureevent$(m \cdot P, u, d)$.

*Proof.* The goal follows from periodicerasurewindow$(T, P, m, N)$. □

**Theorem 3.41.** *For every natural number $T$ and natural number $P$ and natural number $m$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$, *then* recurrentsourceerasure$(m \cdot P, P)$.

*Proof.* The goal follows from periodicrecurrentsourceerasure$(T, P, m)$. □

**Theorem 3.42.** noperiod *if and only if* norecurrentcentralcollision.

*Proof.* The goal follows from noperiodiffnorecurrentcentralcollision. □

**Theorem 3.43.** norowcenterperiod *if and only if* norecurrentcentralcollision.

*Proof.* The goal follows from norowcenterperiodiffnorecurrentcentralcollision. □

**Theorem 3.44.** problemone *if and only if* norecurrentcentralcollision.

*Proof.* The goal follows from problemoneiffnorecurrentcentralcollision. □

**Theorem 3.45.** *If* norecurrentcentralcollision*, then* noperiod*.*

*Proof.* Assume $h$. The goal follows from noperiodofnorecurrentcentralcollision$(h)$. □

**Theorem 3.46.** *For every natural number $T$ and natural number $P$, if $0 < P$, then if* norecurrentcentralcollision*, then not* centerperiodic$(T, P)$*.*

*Proof.* The goal follows from notcenterperiodicofnorecurrentcentralcollision$(T, P)$. □

**Theorem 3.47.** *If* norecurrentcentralcollision*, then* problemone*.*

*Proof.* Assume $h$. The goal follows from noteventuallyperiodicofnorecurrentcentralcollision$(h)$. □

**Theorem 3.48.** *For every natural number $b$ and natural number $d$ and natural number $i$,* pairphase(wittphi$(b)$, wittxor(wittphi$(d)$, wittshl(crosscarry$(b, d), 1)), i) =$ bitvalue(rowbit(wittphi(wittxor$(b, d)), i)) -$ bitvalue(rowbit(wittphi$(b), i)$)*.*

*Proof.* The goal follows from pairphasestep$(b, d, i)$. □

**Theorem 3.49.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* alignedphase$(Q, t, t + k + Q) =$ leftphase$(Q, t, k)$*.*

*Proof.* The goal follows from alignedphaseleft$(Q, t, k)$. □

**Theorem 3.50.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* leftphase$(Q, t, k) = 0$ *if and only if* leftdefect$(Q, t, k) =$ false*.*

*Proof.* The goal follows from phasezeroiff$(Q, t, k)$. □

**Theorem 3.51.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* leftphase$(Q, t, k) = -1$ *or* leftphase$(Q, t, k) = 1$ *if and only if* leftdefect$(Q, t, k) =$ true*.*

*Proof.* The goal follows from phaseunitiff$(Q, t, k)$. □

**Theorem 3.52.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* leftphase$(Q, t+1, k) =$ leftphase$(Q, t, k+1)+$leftphaseflux$(Q, t, k)$*.*

*Proof.* The goal follows from leftphasebalance$(Q, t, k)$. □

**Theorem 3.53.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $\text{collisioncode}(Q, t, k) \neq 0$*, then* $\text{leftphaseflux}(Q, t, k) = -2 \cdot \text{leftphase}(Q, t, k + 1)$.

*Proof.* The goal follows from $\text{collisionphaseflip}(Q, t, k)$. □

**Theorem 3.54.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $0 < k$, then if* $\text{advancingatdepth}(Q, t, k)$*, then* $\text{leftphase}(Q, t + 1, k - 1) = -\text{leftphase}(Q, t, k)$.

*Proof.* The goal follows from $\text{advancingphaseflip}(Q, t, k)$. □

**Theorem 3.55.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* $\text{gatecode}(Q, t, k) = 1 \to \text{leftgatephase}(Q, t, k) = -1$ *and* $(\text{gatecode}(Q, t, k) = 2 \to \text{leftgatephase}(Q, t, k) = -1) \wedge (\text{gatecode}(Q, t, k) = 3 \to \text{leftgatephase}(Q, t, k) = -1) \wedge (\text{gatecode}(Q, t, k) = 4 \to \text{leftgatephase}(Q, t, k) = 1) \wedge (\text{gatecode}(Q, t, k) = 5 \to \text{leftgatephase}(Q, t, k) = 1) \wedge (\text{gatecode}(Q, t, k) = 6 \to \text{leftgatephase}(Q, t, k) = 1)$.

*Proof.* The goal follows from $\text{gatephasecases}(Q, t, k)$. □

**Theorem 3.56.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $\text{erasureevent}(Q, t, k)$*, then* $\text{leftphase}(Q, t + 1, k) = 0$ *and* $\text{leftphaseflux}(Q, t, k) = -\text{leftphase}(Q, t, k + 1) \wedge (\text{leftphase}(Q, t, k + 1) = -1 \vee \text{leftphase}(Q, t, k + 1) = 1)$.

*Proof.* The goal follows from $\text{erasurephasebalance}(Q, t, k)$. □

**Theorem 3.57.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $\text{erasureevent}(Q, t, k)$*, then* $\text{leftcolour}(Q, t + 1, k) = 3 \leftrightarrow \text{leftgatephase}(Q, t, k) \cdot \text{leftphase}(Q, t, k + 1) = -1$ *and* $\text{leftcolour}(Q, t + 1, k) = 0 \leftrightarrow \text{leftgatephase}(Q, t, k) \cdot \text{leftphase}(Q, t, k + 1) = 1$.

*Proof.* The goal follows from $\text{erasureresetproduct}(Q, t, k)$. □

**Theorem 3.58.** *For every natural number $Q$ and natural number $t$ and natural number $k$ and natural number $n$,* $\text{pathphase}(Q, t, k, n) = \text{leftphase}(Q, t, k) + \text{pathflux}(Q, t, k, n)$.

*Proof.* The goal follows from $\text{pathphasebalance}(Q, t, k, n)$. □

**Theorem 3.59.** *For every natural number $Q$ and natural number $t$, if $0 < Q$, then* $\text{leftphase}(Q, t, t + Q) = 1$.

*Proof.* The goal follows from $\text{seedleftphase}(Q, t)$. □

**Theorem 3.60.** *For every natural number $Q$ and natural number $t$, if $0 < Q$, then* $\mathrm{leftphase}(Q, t + (t + Q), 0) = 1 + \mathrm{pathflux}(Q, t, t + Q, t + Q)$.

*Proof.* The goal follows from $\mathrm{seedcenterphasebalance}(Q, t)$. □

**Theorem 3.61.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* $\mathrm{centerperiodic}(T, P)$, *then if $T \leq N$, then* $\mathrm{pathflux}(m \cdot P, N, N + m \cdot P, N + m \cdot P) = -1$.

*Proof.* The goal follows from $\mathrm{periodicseedflux}(T, P, m, N)$. □

**Theorem 3.62.** $\mathrm{leftphase}(3, 0, 3) = 1$ *and* $\mathrm{leftphase}(3, 1, 2) = 0 \wedge \mathrm{leftphaseflux}(3, 0, 2) = -1 \wedge \mathrm{leftgatephase}(3, 0, 2) = 1$.

*Proof.* The goal follows from seedphasecancellation. □

**Theorem 3.63.** *For every natural number $Q$ and natural number $t$ and natural number $k$ and natural number $s$, if* $\mathrm{rightmostleft}(Q, t, k)$, *then if $s < k$, then if* $\mathrm{centercleanthrough}(Q, t, s)$, *then* $\mathrm{rightmostleft}(Q, t + s, k - s)$.

*Proof.* The goal follows from $\mathrm{rightmostleftprofile}(Q, t, k, s)$. □

**Theorem 3.64.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $\mathrm{leadingadvance}(Q, t, k)$, *then if* $\mathrm{centercleanthrough}(Q, t, k)$, *then* $\mathrm{firstpatherasure}(Q, t, k, k - 2)$ *and* $\mathrm{erasureevent}(Q, t + (k - 1), 0) \wedge (\mathrm{gatecode}(Q, t + (k - 1), 0) = 2 \vee \mathrm{gatecode}(Q, t + (k - 1), 0) = 5) \wedge \mathrm{center}(t + (k - 1)) = \mathrm{false}$.

*Proof.* The goal follows from $\mathrm{leadingfirsterasure}(Q, t, k)$. □

**Theorem 3.65.** *For every natural number $Q$ and natural number $b$ and natural number $r$, if $0 < r$, then if* $\mathrm{centerwindowagree}(b, b + Q, r + 1)$, *then if* $\mathrm{center}(b) = \mathrm{true}$, *then if* $\mathrm{leftadjacentdefect}(Q, b + r) = \mathrm{true}$, *then there exists a natural number $s$ such that there exists a natural number $k$ such that $2 \cdot s \leq r + 1$ and $s \leq r \wedge k + s \leq r + 1 \wedge \mathrm{leadingadvance}(Q, b + s, k)$.*

*Proof.* The goal follows from $\mathrm{blackwindowforcesleadingsource}(Q, b, r)$. □

**Theorem 3.66.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* $\mathrm{centerperiodic}(T, P)$, *then if $T \leq N$, then there exists a natural number $t$ such that there exists a natural number $k$ such that $N \leq t$ and $t + 1 \leq 2 \cdot N + m \cdot P + 2 \cdot P \wedge k \leq P \wedge \mathrm{leadingadvance}(m \cdot P, t, k)$.*

*Proof.* The goal follows from periodicleadingsourcewindow$(T, P, m, N)$. □

**Theorem 3.67.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $T \leq N$, then there exists a natural number $t$ such that there exists a natural number $k$ such that $N \leq t$ and $t+1 \leq 2 \cdot N + m \cdot P + 2 \cdot P \wedge k \leq P \wedge$*leadingadvance$(m \cdot P, t, k) \wedge$firstpatherasure$(m \cdot P, t, k, k-2) \wedge (\text{gatecode}(m \cdot P, t + (k-1), 0) = 2 \vee \text{gatecode}(m \cdot P, t + (k-1), 0) = 5)$.

*Proof.* The goal follows from periodicleadingerasurewindow$(T, P, m, N)$. □

**Theorem 3.68.** *For every natural number $T$ and natural number $P$ and natural number $m$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then* recurrentleadingerasure$(m \cdot P, P)$.

*Proof.* The goal follows from periodicrecurrentleadingerasure$(T, P, m)$. □

**Theorem 3.69.** *For every natural number $Q$ and natural number $R$ and natural number $t$ and natural number $k$,* leftphase$(Q+R, t, k) =$ leftphase$(Q, t, k) +$ leftphase$(R, t + Q, k)$.

*Proof.* The goal follows from leftphaselagadd$(Q, R, t, k)$. □

**Theorem 3.70.** *For every natural number $Q$ and natural number $R$ and natural number $t$ and natural number $k$ and natural number $n$,* pathflux$(Q + R, t, k, n) =$ pathflux$(Q, t, k, n) +$ pathflux$(R, t + Q, k, n)$.

*Proof.* The goal follows from pathfluxlagadd$(Q, R, t, k, n)$. □

**Theorem 3.71.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* leadingadvance$(Q, t, k)$*, then if* centercleanthrough$(Q, t, k)$*, then* leftcolour$(Q, t+k, 0) = 3 \leftrightarrow$ leftgatephase$(Q, t+(k-1), 0) \cdot$(leftphase$(Q, t, k) +$ pathflux$(Q, t, k, k-1)) = -1$ *and* leftcolour$(Q, t+k, 0) = 0 \leftrightarrow$ leftgatephase$(Q, t + (k-1), 0) \cdot (\text{leftphase}(Q, t, k) + \text{pathflux}(Q, t, k, k-1)) = 1$.

*Proof.* The goal follows from leadingphasereset$(Q, t, k)$. □

**Theorem 3.72.** *For every natural number $Q$ and natural number $t$, if* leftdefect$(Q, t, 0) =$ false*, then if* leftdefect$(Q, t + 1, 0) =$ false*, then* erasureevent$(Q, t, 0)$ *if and only if* gatecode$(Q, t, 0) = 2$ *or* gatecode$(Q, t, 0) = 5$.

*Proof.* The goal follows from centererasureiffgate$(Q, t)$. □

**Theorem 3.73.** *For every natural number $Q$ and natural number $b$ and natural number $n$ and natural number $i$ and natural number $j$ and natural number $k$, if* commonwhiteblock$(Q, b, n)$*, then if* $i \leq j$*, then if* $j \leq k$*, then if* $k \leq n$*, then if* gatecode$(Q, b + i, 0) = 2$ *or* gatecode$(Q, b + i, 0) = 5$*, then if* gatecode$(Q, b + k, 0) = 2$ *or* gatecode$(Q, b + k, 0) = 5$*, then* gatecode$(Q, b + j, 0) =$ gatecode$(Q, b + i, 0)$.

*Proof.* The goal follows from whiteblockgateconvex$(Q, b, n, i, j, k)$. □

**Theorem 3.74.** *For every natural number $Q$ and natural number $b$ and natural number $n$ and natural number $i$ and natural number $j$ and natural number $k$, if* commonwhiteblock$(Q, b, n)$*, then if* centercleanthrough$(Q, b, n+1)$*, then if* $i \leq j$*, then if* $j \leq k$*, then if* $k \leq n$*, then if* erasureevent$(Q, b+i, 0)$*, then if* erasureevent$(Q, b + k, 0)$*, then* erasureevent$(Q, b + j, 0)$ *and* gatecode$(Q, b + j, 0) =$ gatecode$(Q, b + i, 0)$.

*Proof.* The goal follows from whiteblockerasureconvex$(Q, b, n, i, j, k)$. □

**Theorem 3.75.** leadingadvance$(4, 5, 2)$ *and* centercleanthrough$(4, 5, 2) \wedge$ gatecode$(4, 6, 0) = 5$.

*Proof.* The goal follows from seedleadinggatefive. □

**Theorem 3.76.** leadingadvance$(7, 7, 4)$ *and* centercleanthrough$(7, 7, 4) \wedge$ gatecode$(7, 10, 0) = 2$.

*Proof.* The goal follows from seedleadinggatetwo. □

**Theorem 3.77.** leadingadvance$(11, 4, 4)$ *and* centercleanthrough$(11, 4, 4) \wedge$ gatecode$(11, 7, 0) = 5 \wedge$ leftcolour$(11, 8, 0) = 3$.

*Proof.* The goal follows from seedleadingblackreset. □

**Theorem 3.78.** *For every truth value incomingBit and truth value laterBit and integer incoming and integer gate,* recoverphase$(incomingBit, laterBit,$ transferphase$(incomingBi$ *incoming*.

*Proof.* The goal follows from transferrecover$(incomingBit, laterBit, incoming, gate)$. □

**Theorem 3.79.** *For every truth value incomingBit and truth value laterBit and integer incoming and integer gate,* transferphase$(incomingBit, laterBit, incoming, gate) =$ $0$ *if and only if* $incoming =$ recoverphase$(incomingBit, laterBit, 0, gate)$.

*Proof.* The goal follows from transferkernel($incomingBit, laterBit, incoming, gate$). □

**Theorem 3.80.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* leftphase($Q, t{+}1, k$) = transferphase(leftcell($t, k{+}1$), latergate($Q, t, k$), leftphase($Q, t, k{+}1$), leftgatephase($Q, t, k$)).

*Proof.* The goal follows from phasetransfer($Q, t, k$). □

**Theorem 3.81.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* leftphase($Q, t, k{+}1$) = recoverphase(leftcell($t, k{+}1$), latergate($Q, t, k$), leftphase($Q, t{+}1, k$), leftgatephase($Q, t, k$)).

*Proof.* The goal follows from phaserecover($Q, t, k$). □

**Theorem 3.82.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* erasureevent($Q, t, k$)*, then not* leftgatephase($Q, t, k$) = 0 *and* leftphase($Q, t, k{+}1$) = recoverphase(leftcell($t, k{+}1$), latergate($Q, t, k$), 0, leftgatephase($Q, t, k$)).

*Proof.* The goal follows from erasurerecovery($Q, t, k$). □

**Theorem 3.83.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$ and natural number $g$, if $0 < P$, then if $0 < m$, then if* centerperiodic($T, P$)*, then if $g = 2$ or $g = 5$, then not not there exists a natural number $t$ such that $N \leq t$ and* erasureevent($m \cdot P, t, 0$) $\wedge$ gatecode($m \cdot P, t, 0$) = $g$.

*Proof.* The goal follows from periodicerasureorientation($T, P, m, N, g$). □

**Theorem 3.84.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* leadingadvance($Q, t, k$)*, then if* centercleanthrough($Q, t, k$)*, then there exists a natural number $s$ such that $s + 1 < k$ and* center($t + s$) = true.

*Proof.* The goal follows from leadingblackbeforearrival($Q, t, k$). □

**Theorem 3.85.** *For every natural number $Q$ and natural number $b$ and natural number $r$, if $0 < r$, then if* centerwindowagree($b, b + Q, r + 1$)*, then if* center($b$) = true*, then if* whitecentergap($b, r$)*, then if* leftadjacentdefect($Q, b{+}r$) = true*, then there exists a natural number $k$ such that $k \leq r + 1$ and* leadingadvance($Q, b, k$).

*Proof.* The goal follows from blackgapforcesinitialleading($Q, b, r$). □

**Theorem 3.86.** *For every natural number $Q$ and natural number $t$, if* leadingadvance$(Q, t, 2)$, *then if* centercleanthrough$(Q, t, 2)$, *then* center$(t) =$ true.

*Proof.* The goal follows from leadingdepthtwoblack$(Q, t)$. □

**Theorem 3.87.** *For every natural number $Q$ and natural number $b$ and natural number $r$ and natural number $g$, if $0 < r$, then if* centerwindowagree$(b, b+Q, r+1)$, *then if* center$(b) =$ true, *then if* whitecentergap$(b, r)$, *then if* erasureevent$(Q, b+r, 0)$, *then if* gatecode$(Q, b+r, 0) = g$, *then there exists a natural number $k$ such that $k \leq r+1$ and* leadingadvance$(Q, b, k) \wedge$ firstpatherasure$(Q, b, k, k-2) \wedge$ gatecode$(Q, b+(k-1), 0) = g$.

*Proof.* The goal follows from blackgaperasureorigin$(Q, b, r, g)$. □

**Theorem 3.88.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$ and natural number $g$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$, *then if $g = 2$ or $g = 5$, then not not there exists a natural number $b$ such that there exists a natural number $k$ such that $N \leq b$ and $k \leq P \wedge$* center$(b) =$ true $\wedge$ whitecentergap$(b, k-1) \wedge$ leadingadvance$(m \cdot P, b, k) \wedge$ firstpatherasure$(m \cdot P, b, k, k-2) \wedge$ gatecode$(m \cdot P, b+(k-1), 0) = g$.

*Proof.* The goal follows from periodicblackerasureorigin$(T, P, m, N, g)$. □

**Theorem 3.89.** *For every natural number $t$,* rightcell$(t+1, 1) =$ bxor(center$(t)$, bxor(rightcell$(t, 1)$, right

*Proof.* The goal follows from rightimportstep$(t)$. □

**Theorem 3.90.** *For every natural number $Q$ and natural number $t$, if* center$(t+Q) =$ center$(t)$, *then* rightdefect$(Q, t+1, 1) =$ bxor(rightdefect$(Q, t, 1)$, rightimportdifference$(Q, t)$).

*Proof.* The goal follows from rightdefectimportbalance$(Q, t)$. □

**Theorem 3.91.** *For every natural number $Q$ and natural number $t$, if* center$(t) =$ true, *then if* center$(t+Q) =$ true, *then if* rightdefect$(Q, t, 1) =$ true, *then if* rightdefect$(Q, t+1, 1) =$ true, *then* rightphase$(Q, t+1, 1) = -$ rightphase$(Q, t, 1)$.

*Proof.* The goal follows from blackresetsurvival$(Q, t)$. □

**Theorem 3.92.** *For every natural number $Q$ and natural number $t$, if* center$(t) =$ true, *then if* center$(t+Q) =$ true, *then if* rightdefect$(Q, t, 1) =$ false, *then if* rightdefect$(Q, t+1, 1) =$ true, *then* rightcell$(t, 1) =$ false *and* rightcell$(t+Q, 1) =$ false $\wedge$ rightdefect$(Q, t, 2) =$ true $\wedge$ rightphase$(Q, t+1, 1) = -$ rightphase$(Q, t, 2)$.

*Proof.* The goal follows from blackresetimport$(Q, t)$. □

**Theorem 3.93.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$ and truth value $before$, if $0 < P$, then if $0 < m$, then if* $\mathrm{centerperiodic}(T, P)$*, then not not there exists a natural number $t$ such that $N \leq t$ and* $\mathrm{rightimport}(t) = before \wedge \mathrm{rightimport}(t + m \cdot P) = \mathrm{not}(before) \wedge \mathrm{rightdefect}(m \cdot P, t + 1, 1) = \mathrm{not}(\mathrm{rightdefect}(m \cdot P, t, 1))$.

*Proof.* The goal follows from $\mathrm{periodicrightimportorientation}(T, P, m, N, before)$. □

**Theorem 3.94.** *For every natural number $Q$ and natural number $t$, if* $\mathrm{center}(t + Q) = \mathrm{center}(t)$*, then if* $\mathrm{rightdefect}(Q, t, 1) = \mathrm{false}$*, then if* $\mathrm{rightdefect}(Q, t + 1, 1) = \mathrm{true}$*, then* $\mathrm{rightcell}(t, 1) = \mathrm{false}$ *and* $\mathrm{rightcell}(t+Q, 1) = \mathrm{false} \wedge \mathrm{rightdefect}(Q, t, 2) = \mathrm{true} \wedge \mathrm{rightimportdifference}(Q, t) = \mathrm{true} \wedge \mathrm{rightphase}(Q, t + 1, 1) = (1 - 2 \cdot \mathrm{bitvalue}(\mathrm{center}(t))) \cdot \mathrm{rightphase}(Q, t, 2)$.

*Proof.* The goal follows from $\mathrm{rightheadbirth}(Q, t)$. □

**Theorem 3.95.** *For every natural number $Q$ and natural number $start$ and natural number $finish$, if* $\mathrm{headimportpath}(Q, start, finish)$*, then if* $\mathrm{centercleanthrough}(Q, start, finish - start)$*, then* $\mathrm{rightphase}(Q, finish, 1) = \mathrm{centersign}(start, finish - start) \cdot \mathrm{rightphase}(Q, start, 2)$.

*Proof.* The goal follows from $\mathrm{headimportpathphase}(Q, start, finish)$. □

**Theorem 3.96.** *For every natural number $Q$ and natural number $t$, if* $\mathrm{leftdefect}(Q, t, 0) = \mathrm{false}$*, then if* $\mathrm{leftdefect}(Q, t + 1, 0) = \mathrm{false}$*, then if* $\mathrm{center}(t) = \mathrm{false}$*, then* $\mathrm{leftphase}(Q, t, 1) = (1 - 2 \cdot \mathrm{bitvalue}(\mathrm{center}(t + 1))) \cdot \mathrm{rightphase}(Q, t, 1)$.

*Proof.* The goal follows from $\mathrm{whitecenterphasematching}(Q, t)$. □

**Theorem 3.97.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$ and natural number $g$, if $0 < P$, then if $0 < m$, then if* $\mathrm{centerperiodic}(T, P)$*, then if $g = 2$ or $g = 5$, then not not there exists a natural number $start$ such that there exists a natural number $b$ such that there exists a natural number $k$ such that $N \leq start$ and $N \leq b \wedge k \leq P \wedge \mathrm{center}(b) = \mathrm{true} \wedge \mathrm{whitecentergap}(b, k - 1) \wedge \mathrm{leadingadvance}(m \cdot P, b, k) \wedge \mathrm{firstpatherasure}(m \cdot P, b, k, k - 2) \wedge \mathrm{gatecode}(m \cdot P, b + (k - 1), 0) = g \wedge \mathrm{headimportpath}(m \cdot P, start, b + (k - 1)) \wedge \mathrm{leftphase}(m \cdot P, b, k) + \mathrm{pathflux}(m \cdot P, b, k, k - 1) = (1 - 2 \cdot \mathrm{bitvalue}(\mathrm{center}(b + k))) \cdot \mathrm{centersign}(start, b + (k - 1) - start) \cdot \mathrm{rightphase}(m \cdot P, start, 2)$.

*Proof.* The goal follows from $\mathrm{periodiccoupledcancellation}(T, P, m, N, g)$. □

**Theorem 3.98.** *For every natural number $Q$ and natural number $b$ and natural number $k$, if* $\mathrm{leadingadvance}(Q, b, k)$*, then if* $\mathrm{centercleanthrough}(Q, b, k)$*, then* $\mathrm{leftphase}(Q, b, k) + \mathrm{pathflux}(Q, b, k, k-1) = (1 - 2 \cdot \mathrm{bitvalue}(\mathrm{center}(b + k))) \cdot \mathrm{leftgatephase}(Q, b + (k-1), 0)$.

*Proof.* The goal follows from $\mathrm{leadingarrivalphaseeqgatephase}(Q, b, k)$. □

**Theorem 3.99.** *For every natural number $Q$ and natural number $start$ and natural number $finish$, if* $\mathrm{headimportpath}(Q, start, finish)$*, then if* $\mathrm{centercleanthrough}(Q, start, finish - start)$*, then if* $\mathrm{center}(finish) = \mathrm{false}$*, then* $\mathrm{leftgatephase}(Q, finish, 0) = \mathrm{centersign}(start, finish - start) \cdot \mathrm{rightphase}(Q, start, 2)$.

*Proof.* The goal follows from $\mathrm{headimportpathgatephase}(Q, start, finish)$. □

**Theorem 3.100.** *For every natural number $Q$ and natural number $start$ and natural number $finish$, if* $\mathrm{headimportpath}(Q, start, finish)$*, then if* $\mathrm{centercleanthrough}(Q, start, finish - start)$*, then if* $\mathrm{center}(finish) = \mathrm{false}$*, then not* $\mathrm{leftgatephase}(Q, finish, 0) = -(\mathrm{centersign}(start, finish - start) \cdot \mathrm{rightphase}(Q, start, 2))$.

*Proof.* The goal follows from $\mathrm{headimportpathrejectsnegativegate}(Q, start, finish)$. □

**Theorem 3.101.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$ and natural number $g$, if $0 < P$, then if $0 < m$, then if* $\mathrm{centerperiodic}(T, P)$*, then if $g = 2$ or $g = 5$, then not not there exists a natural number start such that there exists a natural number b such that there exists a natural number k such that $N \leq start$ and $N \leq b \wedge k \leq P \wedge \mathrm{center}(b) = \mathrm{true} \wedge \mathrm{whitecentergap}(b, k-1) \wedge \mathrm{leadingadvance}(m \cdot P, b, k) \wedge \mathrm{firstpatherasure}(m \cdot P, b, k, k-2) \wedge \mathrm{gatecode}(m \cdot P, b + (k-1), 0) = g \wedge \mathrm{headimportpath}(m \cdot P, start, b + (k-1)) \wedge \mathrm{leftgatephase}(m \cdot P, b + (k-1), 0) = \mathrm{centersign}(start, b + (k-1) - start) \cdot \mathrm{rightphase}(m \cdot P, start, 2)$.

*Proof.* The goal follows from $\mathrm{periodiccoupledgatephasesign}(T, P, m, N, g)$. □

**Theorem 3.102.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* $\mathrm{centerperiodic}(T, P)$*, then if $T \leq N$, then* $\mathrm{pathflux}(m \cdot P + m \cdot P, N, N + (m \cdot P + m \cdot P), N + (m \cdot P + m \cdot P)) = -1$ *and* $\mathrm{pathflux}(m \cdot P, N, N + (m \cdot P + m \cdot P), N + (m \cdot P + m \cdot P)) = 0 \wedge \mathrm{pathflux}(m \cdot P, N + m \cdot P, N + (m \cdot P + m \cdot P), N + (m \cdot P + m \cdot P)) = -1$.

*Proof.* The goal follows from periodicdoubledseedfluxsplit$(T, P, m, N)$. □

**Theorem 3.103.** leadingadvance$(11, 4, 4)$ *and* centercleanthrough$(11, 4, 4) \wedge$ headimportpath$(11, 6, 7) \wedge$centercleanthrough$(11, 6, 1) \wedge$leftgatephase$(11, 7, 0) = 1 \wedge$ centersign$(6, 1) = 1 \wedge$ rightphase$(11, 6, 2) = 1$.

*Proof.* The goal follows from seedcoupledgatesign. □

**Theorem 3.104.** *For every natural number $Q$ and natural number $t$ and natural number $j$, if* rightdefect$(Q, t+1, j+1) =$ true*, then* rightdefect$(Q, t, j) =$ true *or* rightdefect$(Q, t, j + 1) =$ true $\vee$ rightdefect$(Q, t, j + 2) =$ true.

*Proof.* The goal follows from rightdefecthasancestor$(Q, t, j)$. □

**Theorem 3.105.** *For every natural number $Q$ and natural number $t$ and natural number $j$ and natural number $T$, if $1 \leq j$, then if $T \leq t$, then if* rightdefect$(Q, t, j) =$ true*, then if* centercleanthrough$(Q, T, t - T)$*, then* rightlineage$(Q, t, j, T)$.

*Proof.* The goal follows from rightlineageofcleancenter$(Q, t, j, T)$. □

**Theorem 3.106.** *For every natural number $Q$ and natural number $t$ and natural number $k$ and natural number $T$, if $1 \leq k$, then if $T \leq t$, then if* leftdefect$(Q, t, k) =$ true*, then if* centercleanthrough$(Q, T, t - T)$*, then* leftlineage$(Q, t, k, T)$.

*Proof.* The goal follows from leftlineageofcleancenter$(Q, t, k, T)$. □

**Theorem 3.107.** *For every natural number $Q$ and natural number $t$ and natural number $j$ and natural number $T$, if $T \leq t$, then if* rightlineage$(Q, t, j, T)$*, then there exists a natural number $i$ such that $1 \leq i$ and $i \leq j + (t - T) \wedge j \leq i + (t - T) \wedge$* rightdefect$(Q, T, i) =$ true.

*Proof.* The goal follows from rightlineageatstart$(Q, t, j, T)$. □

**Theorem 3.108.** *For every natural number $Q$ and natural number $t$ and natural number $k$ and natural number $T$, if $T \leq t$, then if* leftlineage$(Q, t, k, T)$*, then there exists a natural number $i$ such that $1 \leq i$ and $i \leq k + (t - T) \wedge k \leq i + (t - T) \wedge$* leftdefect$(Q, T, i) =$ true.

*Proof.* The goal follows from leftlineageatstart$(Q, t, k, T)$. □

**Theorem 3.109.** *For every natural number $Q$ and natural number $t$ and natural number $j$, if* $\mathrm{rightlineage}(Q, t, j, 0)$*, then there exists a natural number $i$ such that $1 \leq i$ and $i \leq Q \wedge i \leq j + t \wedge j \leq i + t \wedge \mathrm{rightdefect}(Q, 0, i) = \mathrm{true} \wedge \mathrm{rightcell}(Q, i) = \mathrm{true} \wedge \mathrm{rightphase}(Q, 0, i) = 1$.*

*Proof.* The goal follows from $\mathrm{rightlineageseed}(Q, t, j)$. □

**Theorem 3.110.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $\mathrm{leftlineage}(Q, t, k, 0)$*, then there exists a natural number $i$ such that $1 \leq i$ and $i \leq Q \wedge i \leq k + t \wedge k \leq i + t \wedge \mathrm{leftdefect}(Q, 0, i) = \mathrm{true} \wedge \mathrm{leftcell}(Q, i) = \mathrm{true} \wedge \mathrm{leftphase}(Q, 0, i) = 1$.*

*Proof.* The goal follows from $\mathrm{leftlineageseed}(Q, t, k)$. □

**Theorem 3.111.** *For every natural number $Q$ and natural number $t$, if $0 < Q$, then there exists a natural number $y$ such that $y \leq t + Q$ and* $\mathrm{rightfront}(Q, t, y)$*.*

*Proof.* The goal follows from $\mathrm{existsrightfront}(Q, t)$. □

**Theorem 3.112.** *For every natural number $Q$ and natural number $t$ and natural number $y$ and natural number $z$, if* $\mathrm{leftdefect}(Q, t, 0) = \mathrm{false}$*, then if* $\mathrm{rightfront}(Q, t, y)$*, then if* $\mathrm{rightfront}(Q, t + 1, z)$*, then $y \leq z + 1$.*

*Proof.* The goal follows from $\mathrm{rightfrontlower}(Q, t, y, z)$. □

**Theorem 3.113.** *For every natural number $Q$ and natural number $t$ and natural number $y$, if* $\mathrm{leftdefect}(Q, t, 0) = \mathrm{false}$*, then if* $\mathrm{rightfront}(Q, t, y)$*, then if $2 \leq y$, then* $\mathrm{rightdefect}(Q, t + 1, y - 1) = \mathrm{not}(\mathrm{rightcell}(t, y - 1))$*.*

*Proof.* The goal follows from $\mathrm{rightfrontdescent}(Q, t, y)$. □

**Theorem 3.114.** *For every natural number $Q$ and natural number $t$ and natural number $y$ and natural number $z$, if* $\mathrm{leftdefect}(Q, t, 0) = \mathrm{false}$*, then if* $\mathrm{rightfront}(Q, t, y)$*, then if $2 \leq y$, then if* $\mathrm{rightfront}(Q, t + 1, z)$*, then if $z + 1 \leq y$, then $z = y - 1$ and* $\mathrm{rightcell}(t, y - 1) = \mathrm{false}$*.*

*Proof.* The goal follows from $\mathrm{rightfrontlockstep}(Q, t, y, z)$. □

**Theorem 3.115.** *For every natural number $Q$ and natural number $b$ and natural number $k$ and natural number $s$, if* $\mathrm{rightmostleft}(Q, b, k)$*, then if* $\mathrm{centercleanthrough}(Q, b, k)$*, then if $s+1 < k$, then if* $\mathrm{rightdefect}(Q, b+s, 1) = \mathrm{true}$*, then* $\mathrm{center}(b + s) = \mathrm{true}$*.*

*Proof.* The goal follows from $\mathrm{leadingheadblackbeforearrival}(Q, b, k, s)$. □

**Theorem 3.116.** *For every natural number $Q$ and natural number $b$ and natural number $k$ and natural number start, if $3 \leq k$, then if* leadingadvance$(Q, b, k)$*, then if* centercleanthrough$(Q, b, k)$*, then if* whitecentergap$(b, k-1)$*, then if* headimportpath$(Q, start, b + (k-1))$*, then $start = b + (k-2)$.*

*Proof.* The goal follows from coupledimportbornatarrival$(Q, b, k, start)$. □

**Theorem 3.117.** *For every natural number $Q$ and natural number $b$ and natural number $k$ and natural number $s$, if $0 < Q$, then if $3 \leq k$, then if* leadingadvance$(Q, b, k)$*, then if* centercleanthrough$(Q, b, k)$*, then if* whitecentergap$(b, k-1)$*, then if $1 \leq s$, then if $s+1 < k$, then there exists a natural number $y$ such that* rightdefect$(Q, b+s, 1) =$ false *and* rightfront$(Q, b+s, y) \wedge 2 \leq y \wedge y \leq k - s$.

*Proof.* The goal follows from coupledrightfrontmirrorat$(Q, b, k, s)$. □

**Theorem 3.118.** *For every natural number $Q$ and natural number $b$ and natural number $k$ and natural number $s$, if $0 < Q$, then if $3 \leq k$, then if* leadingadvance$(Q, b, k)$*, then if* centercleanthrough$(Q, b, k)$*, then if* whitecentergap$(b, k-1)$*, then if $1 \leq s$, then if $s + 2 < k$, then if* rightfront$(Q, b+s, k-s)$*, then* rightcell$(b+s, k-s-1) =$ false *and* rightfront$(Q, b+(s+1), k-(s+1))$.

*Proof.* The goal follows from coupledwhitestaircaseat$(Q, b, k, s)$. □

**Theorem 3.119.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$ and natural number $g$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $g = 2$ or $g = 5$, then not not there exists a natural number start such that there exists a natural number $b$ such that there exists a natural number $k$ such that there exists a natural number $y$ such that $N \leq start$ and $N \leq b \wedge T \leq b \wedge T \leq start \wedge 2 \leq k \wedge k \leq P \wedge$* center$(b) =$ true $\wedge$ whitecentergap$(b, k-1) \wedge$ leadingadvance$(m \cdot P, b, k) \wedge$ gatecode$(m \cdot P, b+(k-1), 0) = g \wedge$ headimportpath$(m \cdot P, start, b+(k-1)) \wedge (3 \leq k \to start = b+(k-2)) \wedge (3 \leq k \to$ rightfront$(m \cdot P, b+1, y) \wedge 2 \leq y \wedge y \leq k-1) \wedge$ rightlineage$(m \cdot P, start, 2, T) \wedge$ leftlineage$(m \cdot P, b, k, T)$.

*Proof.* The goal follows from periodiccoupledimportancestry$(T, P, m, N, g)$. □

**Theorem 3.120.** leadingadvance$(14, 5, 3)$ *and* centercleanthrough$(14, 5, 3) \wedge$ whitecentergap$(5, 2) \wedge$center$(5) =$ true $\wedge$ rightfront$(14, 6, 2) \wedge$headimportpath$(14, 6, 7)$.

*Proof.* The goal follows from seedcoupledimportatarrival. □

**Theorem 3.121.** *For every natural number $t$ and natural number $k$ and natural number $n$,* $\mathrm{bitvalue}(\mathrm{raybit}(t,k,n)) = \mathrm{bitvalue}(\mathrm{raybit}(t,k,0)) + \mathrm{rayflux}(t,k,n)$.

*Proof.* The goal follows from $\mathrm{raybitbalance}(t,k,n)$. □

**Theorem 3.122.** *For every natural number $Q$ and natural number $d$ and natural number $t$ and natural number $p$, if $d+p \leq t$, then* $\mathrm{leftphase}(Q+Q, t+(t-d-p), p) = \mathrm{diagonalphase}(Q,d,t) + \mathrm{diagonalflux}(Q,d,t,p)$.

*Proof.* The goal follows from $\mathrm{diagonalraybalance}(Q,d,t,p)$. □

**Theorem 3.123.** *For every natural number $Q$ and natural number $R$ and natural number $d$ and natural number $t$ and natural number $p$,* $\mathrm{diagonalflux}(Q+R,d,t,p) = \mathrm{diagonalflux}(Q,d,t,p) + \mathrm{diagonalflux}(R,d,t+Q,p)$.

*Proof.* The goal follows from $\mathrm{diagonalfluxlagadd}(Q,R,d,t,p)$. □

**Theorem 3.124.** *For every natural number $Q$ and natural number $z$ and natural number $t$, if* $\mathrm{zerodiagonal}(z)$*, then if* $\mathrm{doublesat}(Q,z)$*, then if* $\mathrm{settledby}(z+1,t)$*, then* $\mathrm{diagonalphase}(Q,z+1,t) = -1 \vee \mathrm{diagonalphase}(Q,z+1,t) = 1$ *and* $\mathrm{diagonalphase}(Q,z+1,t+Q) = -\mathrm{diagonalphase}(Q,z+1,t) \wedge \mathrm{diagonalphase}(Q+Q,z+1,t) = 0$.

*Proof.* The goal follows from $\mathrm{diagonalphasedoubles}(Q,z,t)$. □

**Theorem 3.125.** *For every natural number $Q$ and natural number $z$ and natural number $t$, if* $\mathrm{zerodiagonal}(z)$*, then if* $\mathrm{settledby}(z+2,t)$*, then* $\mathrm{diagonalphase}(Q,z+3,t+1) = -\mathrm{diagonalphase}(Q,z+1,t)$.

*Proof.* The goal follows from $\mathrm{diagonalresetphase}(Q,z,t)$. □

**Theorem 3.126.** *For every natural number $Q$ and natural number $z$ and natural number $t$, if* $\mathrm{zerodiagonal}(z)$*, then if* $\mathrm{settledby}(z+2,t)$*, then if* $z+3 \leq t$*, then* $\mathrm{diagonalflux}(Q,z+3,t+1,1) = \mathrm{diagonalflux}(Q,z+1,t,1) + 2 \cdot \mathrm{diagonalphase}(Q,z+1,t)$.

*Proof.* The goal follows from $\mathrm{diagonalresetfluxjump}(Q,z,t)$. □

**Theorem 3.127.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $d$ and natural number $t$, if* $\mathrm{centerperiodic}(T,P)$*, then if* $d+1 \leq t$*, then if* $T \leq t+(t-d-1)$*, then if* $\mathrm{center}(t+(t-d-1)) = \mathrm{false}$*, then* $\mathrm{diagonalphase}(m \cdot P, d, t) + \mathrm{diagonalflux}(m \cdot P, d, t, 1) = (1 - 2 \cdot \mathrm{bitvalue}(\mathrm{center}(t+(t-d-1)+1))) \cdot \mathrm{rightphase}(m \cdot P + m \cdot P, t+(t-d-1), 1)$.

*Proof.* The goal follows from periodicdiagonalwhiteheadbalance$(T, P, m, d, t)$. □

**Theorem 3.128.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $d$ and natural number $t$ and natural number start, if* centerperiodic$(T, P)$*, then if* $d + 1 \leq t$*, then if* $T \leq start$*, then if* headimportpath$(m \cdot P + m \cdot P, start, t + (t - d - 1))$*, then if* center$(t + (t - d - 1)) =$ false*, then* diagonalphase$(m \cdot P, d, t) +$ diagonalflux$(m \cdot P, d, t, 1) = (1 - 2 \cdot$ bitvalue(center$(t + (t - d - 1) + 1))) \cdot ($centersign$(start, t + (t - d - 1) - start) \cdot$ rightphase$(m \cdot P + m \cdot P, start, 2))$.

*Proof.* The goal follows from periodicdiagonalimportbalance$(T, P, m, d, t, start)$. □

**Theorem 3.129.** *For every natural number $Q$ and natural number $b$ and natural number $k$ and natural number start and natural number $q$, if* $0 < Q$*, then if* $3 \leq k$*, then if* leadingadvance$(Q, b, k)$*, then if* centercleanthrough$(Q, b, k)$*, then if* whitecentergap$(b, k-1)$*, then if* headimportpath$(Q, start, b+(k-1))$*, then if* $1 \leq q$*, then if* $q + 1 < k$*, then* rightfront$(Q, b + 1, k - 1)$ *and* rightcell$(b + 1, q) =$ false $\wedge$ rightcell$(b + 1 + Q, q) =$ false*.*

*Proof.* The goal follows from coupledrightfrontexact$(Q, b, k, start, q)$. □

**Theorem 3.130.** *For every natural number $Q$ and natural number $b$ and natural number $k$ and natural number start and natural number $i$, if* $0 < Q$*, then if* $3 \leq k$*, then if* leadingadvance$(Q, b, k)$*, then if* centercleanthrough$(Q, b, k)$*, then if* whitecentergap$(b, k - 1)$*, then if* headimportpath$(Q, start, b + (k - 1))$*, then if* $1 \leq i$*, then if* $i + 1 < k$*, then* rightfront$(Q, b + i, k - i)$ *and* rightcell$(b + i, k - i - 1) =$ false $\wedge$ rightcell$(b + i + Q, k - i - 1) =$ false*.*

*Proof.* The goal follows from coupledforcedwhitestaircase$(Q, b, k, start, i)$. □

**Theorem 3.131.** *For every natural number $Q$ and natural number $b$ and natural number start, if* leadingadvance$(Q, b, 2)$*, then if* centercleanthrough$(Q, b, 2)$*, then if* headimportpath$(Q, start, b+1)$*, then* center$(b) =$ true *and* center$(b+1) =$ false $\wedge$($start < b \wedge$ rightdefect$(Q, b, 1) =$ true $\wedge$ rightimportdifference$(Q, b) =$ false $\wedge$ leftphase$(Q, b, 2) = (1-2 \cdot$ bitvalue(center$(b+2))) \cdot$ rightphase$(Q, b, 1) \vee$ $start = b \wedge$ rightdefect$(Q, b, 1) =$ false $\wedge$ rightcell$(b, 1) =$ false $\wedge$ rightcell$(b + Q, 1) =$ false $\wedge$ rightdefect$(Q, b, 2) =$ true $\wedge$ rightimportdifference$(Q, b) =$ true $\wedge$ leftphase$(Q, b, 2) = (1-2 \cdot$ bitvalue(center$(b+2))) \cdot$ rightphase$(Q, b, 2))$.

*Proof.* The goal follows from depthtwocancellationcases$(Q, b, start)$. □

**Theorem 3.132.** *For every natural number $Q$ and natural number $b$ and natural number $k$ and natural number start, if* leadingadvance$(Q, b, k)$*, then if* centercleanthrough$(Q, b, k)$*, then if* whitecentergap$(b, k-1)$*, then if* headimportpath$(Q, start, b+(k-1))$*, then start* $= b+(k-2)$ *or* $k = 2$ *and start* $< b \wedge$ rightdefect$(Q, b, 1) =$ true.

*Proof.* The goal follows from coupledimportbirthordepthtwosurvival$(Q, b, k, start)$. □

**Theorem 3.133.** *For every natural number $Q$ and natural number $b$ and natural number $k$ and natural number start and natural number $q$, if* $0 < Q$*, then if* $3 \leq k$*, then if* leadingadvance$(Q, b, k)$*, then if* centercleanthrough$(Q, b, k)$*, then if* center$(b) =$ true*, then if* whitecentergap$(b, k-1)$*, then if* headimportpath$(Q, start, b+(k-1))$*, then if* $q+2 < k$*, then* rightcell$(b, q) =$ true *and* rightcell$(b+Q, q) =$ true.

*Proof.* The goal follows from coupledblackrootprefix$(Q, b, k, start, q)$. □

**Theorem 3.134.** leadingadvance$(4, 5, 2)$ *and* centercleanthrough$(4, 5, 2) \wedge$ headimportpath$(4, 2, 6) \wedge$ rightdefect$(4, 5, 1) =$ true $\wedge$ rightimportdifference$(4, 5) =$ false.

*Proof.* The goal follows from seeddepthtwosurvival. □

**Theorem 3.135.** *For every natural number $Q$ and natural number $t$,* alignedbackground$(Q, t+1) =$ wittphi(alignedbackground$(Q, t)$).

*Proof.* The goal follows from alignedbackgroundstep$(Q, t)$. □

**Theorem 3.136.** *For every natural number $Q$ and natural number $t$,* aligneddefect$(Q, t+1) =$ defectstep(alignedbackground$(Q, t)$, aligneddefect$(Q, t)$).

*Proof.* The goal follows from aligneddefectstep$(Q, t)$. □

**Theorem 3.137.** *For every natural number $Q$ and natural number $t$,* rowbit(alignedbackground$(Q, t), 2 \cdot t + Q) =$ true.

*Proof.* The goal follows from backgroundtopbit$(Q, t)$. □

**Theorem 3.138.** *For every natural number $Q$ and natural number $t$, if* $0 < Q$*, then* rowbit(aligneddefect$(Q, t), 2 \cdot (t + Q)) =$ true.

*Proof.* The goal follows from aligneddefecttopbit$(Q, t)$. □

**Theorem 3.139.** *For every natural number $Q$ and natural number $t$, if* $0 < Q$, *then* $\mathrm{twopow}(2 \cdot (t + Q)) \leq \mathrm{aligneddefect}(Q, t)$ *and* $\mathrm{aligneddefect}(Q, t) < \mathrm{twopow}(2 \cdot (t + Q) + 1)$.

*Proof.* The goal follows from $\mathrm{aligneddefectmagnitude}(Q, t)$. □

**Theorem 3.140.** *For every natural number $Q$ and natural number $t$ and natural number $j$, if* $2 \cdot (t+Q) < j$, *then* $\mathrm{rowbit}(\mathrm{aligneddefect}(Q, t), j) = \mathrm{false}$.

*Proof.* The goal follows from $\mathrm{aligneddefectabovetop}(Q, t, j)$. □

**Theorem 3.141.** *For every natural number $Q$ and natural number $t$ and natural number $j$, if* $j < Q$, *then* $\mathrm{rowbit}(\mathrm{aligneddefect}(Q, t), j) = \mathrm{rowbit}(\mathrm{wittrow}(t+ Q), j)$.

*Proof.* The goal follows from $\mathrm{aligneddefectlowprefix}(Q, t, j)$. □

**Theorem 3.142.** *For every natural number $b$ and natural number $d$,* $\mathrm{defectstep}(b, d) = \mathrm{wittxor}(\mathrm{wittlinear}(d), \mathrm{wittshl}(\mathrm{wittxor}(\mathrm{wittcarry}(d), \mathrm{crosscarry}(b, d)), 1))$.

*Proof.* The goal follows from $\mathrm{defectsteplinear}(b, d)$. □

**Theorem 3.143.** *For every natural number $b$ and natural number $d$,* $\mathrm{rowbit}(\mathrm{defectstep}(b, d), 0) = \mathrm{rowbit}(d, 0)$.

*Proof.* The goal follows from $\mathrm{defectsteplowbit}(b, d)$. □

**Theorem 3.144.** *For every natural number $b$ and natural number $d$,* $\mathrm{vala}(\mathrm{defectstep}(b, d)) = \mathrm{valb}(\mathrm{wittxor}(\mathrm{wittcarry}(d), \mathrm{crosscarry}(b, d)))$ *and* $\mathrm{valb}(\mathrm{defectstep}(b, d)) = \mathrm{bxor}(\mathrm{vala}(\mathrm{wittxor}(\mathrm{wittcarry}(d$

*Proof.* The goal follows from $\mathrm{defectstepvalcoordinates}(b, d)$. □

**Theorem 3.145.** *For every natural number $j$,* $\mathrm{vala}(\mathrm{wittshl}(1, j + 3)) = \mathrm{vala}(\mathrm{wittshl}(1, j))$ *and* $\mathrm{valb}(\mathrm{wittshl}(1, j+3)) = \mathrm{valb}(\mathrm{wittshl}(1, j)) \wedge (\mathrm{vala}(\mathrm{wittshl}(1, j)) = \mathrm{true} \vee \mathrm{valb}(\mathrm{wittshl}(1, j)) = \mathrm{true})$.

*Proof.* The goal follows from $\mathrm{edgevalcoordinates}(j)$. □

**Theorem 3.146.** *For every natural number $t$ and natural number $k$,* $\mathrm{aligneddefect}(\mathrm{twopow}(k), t) = \mathrm{wittxor}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), \mathrm{wittxor}(\mathrm{wittrow}(t), \mathrm{wittshl}(\mathrm{wittrow}(t), \mathrm{twopow}(k)+ \mathrm{twopow}(k))))$.

*Proof.* The goal follows from $\mathrm{aligneddefectwindow}(t, k)$. □

**Theorem 3.147.** *For every natural number $t$ and natural number $k$, if* $t <$ twopow$(k)$*, then* rowbit(aligneddefect(twopow$(k), t), 0)$ = true *and* rowbit(wittwin$(t,$ twopow$(k)), 0) =$ false $\wedge$ rowbit(aligneddefect(twopow$(k), t), 2\cdot(t+$twopow$(k))) =$ true $\wedge$ rowbit(wittwin$(t,$ twopow$(k)), 2\cdot$ $(t +$ twopow$(k))) =$ false $\wedge$ vala(wittwin$(t,$ twopow$(k))) =$ vala(wittrow$(t +$ twopow$(k)))\wedge$valb(wittwin$(t,$ twopow$(k))) =$ valb(wittrow$(t+$twopow$(k)))\wedge$ rowbit(aligneddefect(twopow$(k), t), t+$twopow$(k)) =$ rowbit(wittwin$(t,$ twopow$(k)), t+$ twopow$(k))$.

*Proof.* The goal follows from dyadicwindowreadings$(t, k)$. □

**Theorem 3.148.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if* centerperiodic$(T, P)$*, then if* $T \leq$ $t$*, then* rowbit(wittphi(aligneddefect$(m\cdot P, t)), t+m\cdot P+1) =$ rowbit(crosscarry(alignedbackground$(m\cdot$ $P, t),$ aligneddefect$(m \cdot P, t)), t + m \cdot P)$.

*Proof.* The goal follows from periodiccentercrosscarry$(T, P, m, t)$. □

**Theorem 3.149.** aligneddefect$(1, 0) = 5$ *and* vala(aligneddefect$(1, 0)) =$ false $\wedge$ valb(aligneddefect$(1, 0)) =$ true $\wedge$ aligneddefect$(1, 2) = 93\wedge$vala(aligneddefect$(1, 2)) =$ false $\wedge$ valb(aligneddefect$(1, 2)) =$ false $\wedge$ rowbit(aligneddefect$(1, 2), 3) =$ true $\wedge$ aligneddefect$(4, 0) =$ $385\wedge$vala(aligneddefect$(4, 0)) =$ false $\wedge$ valb(aligneddefect$(4, 0)) =$ false $\wedge$ rowbit(aligneddefect$(4, 0), 4) =$ false.

*Proof.* The goal follows from seedvalcancellation. □

**Theorem 3.150.** *For every natural number $d$ and natural number $n$ and natural number $j$, if $j < n$ or $n+2 < j$, then* rowbit(wittxor$(d,$ centerflipmask$(n)), n) =$ not(rowbit$(d, n))$ *and* rowbit(wittxor$(d,$ centerflipmask$(n)), j) =$ rowbit$(d, j)\wedge$ vala(wittxor$(d,$ centerflipmask$(n))) =$ vala$(d)\wedge$valb(wittxor$(d,$ centerflipmask$(n))) =$ valb$(d)$.

*Proof.* The goal follows from centerflipmaskinvisible$(d, n, j)$. □

**Theorem 3.151.** *For every natural number $t$ and natural number $j$,* rowbit(observercode$(t), j) =$ leftcell$(t, j)$.

*Proof.* The goal follows from observercodereadout$(t, j)$. □

**Theorem 3.152.** *For every natural number $Q$ and natural number $t$,* observerdefect$(Q, t) =$ wittxor(observercode$(t + Q),$ observercode$(t))$.

*Proof.* The goal follows from observerdefectalignment$(Q, t)$. □

**Theorem 3.153.** *For every natural number $Q$ and natural number $t$ and natural number $j$,* $\mathrm{rowbit}(\mathrm{observerdefect}(Q,t),j) = \mathrm{rowbit}(\mathrm{aligneddefect}(Q,t),t+Q+j)$.

*Proof.* The goal follows from $\mathrm{observerdefectreadout}(Q,t,j)$. □

**Theorem 3.154.** *For every natural number $t$,* $\mathrm{vala}(\mathrm{observercode}(t+1)) = \mathrm{bxor}(\mathrm{center}(t),\mathrm{bxor}(\mathrm{vala}(\mathrm{wittcarry}(\mathrm{observercode}(t))),\mathrm{observerinput}(t)))$ *and* $\mathrm{valb}(\mathrm{observercode}(t+1)) = \mathrm{bxor}(\mathrm{center}(t),\mathrm{valb}(\mathrm{wittcarry}(\mathrm{observercode}(t))))$.

*Proof.* The goal follows from $\mathrm{observervalstep}(t)$. □

**Theorem 3.155.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if* $\mathrm{centerperiodic}(T,P)$*, then if* $T \leq t$*, then* $\mathrm{vala}(\mathrm{observerdefect}(m\cdot P,t+1)) = \mathrm{bxor}(\mathrm{vala}(\mathrm{wittxor}(\mathrm{wittcarry}(\mathrm{observercode}(t+m\cdot P)),\mathrm{wittcarry}(\mathrm{observercode}(t)))),\mathrm{leftdefect}(m\cdot P,t,1))$ *and* $\mathrm{valb}(\mathrm{observerdefect}(m\cdot P,t+1)) = \mathrm{valb}(\mathrm{wittxor}(\mathrm{wittcarry}(\mathrm{observercode}(t+m\cdot P)),\mathrm{wittcarry}(\mathrm{observercode}(t))))$.

*Proof.* The goal follows from $\mathrm{periodicobservervalstep}(T,P,m,t)$. □

**Theorem 3.156.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if* $0 < P$*, then if* $0 < m$*, then if* $\mathrm{centerperiodic}(T,P)$*, then if* $T \leq t$*, then* $\mathrm{rowbit}(\mathrm{observerdefect}(m\cdot P,t),t+m\cdot P) = \mathrm{true}$ *and* $\mathrm{rowbit}(\mathrm{observerdefect}(m\cdot P,t),0) = \mathrm{false}$.

*Proof.* The goal follows from $\mathrm{periodicobserveredges}(T,P,m,t)$. □

**Theorem 3.157.** $\mathrm{observerdefect}(2,0) = 7$ *and* $\mathrm{vala}(\mathrm{observerdefect}(2,0)) = \mathrm{false} \wedge \mathrm{valb}(\mathrm{observerdefect}(2,0)) = \mathrm{false} \wedge \mathrm{rowbit}(\mathrm{observerdefect}(2,0),0) = \mathrm{true} \wedge \mathrm{observerdefect}(4,0) = 24 \wedge \mathrm{vala}(\mathrm{observerdefect}(4,0)) = \mathrm{true} \wedge \mathrm{valb}(\mathrm{observerdefect}(4,0)) = \mathrm{true} \wedge \mathrm{rowbit}(\mathrm{observerdefect}(4,0),0) = \mathrm{false}$.

*Proof.* The goal follows from $\mathrm{observerseedzeroval}$. □

# 4 Cyclotomic observer transport

**Theorem 4.1.** *For every natural number $r$ and natural number $a$,* $\mathrm{qdivides}(r,\mathrm{wittlinpow}(r,a))$.

*Proof.* The goal follows from $\mathrm{qdividesintro}(r,a)$. □

**Theorem 4.2.** *For every natural number $r$ and natural number $d$, if* $\mathrm{qdivides}(r,d)$*, then there exists a natural number $a$ such that* $\mathrm{wittlinpow}(r,a) = d$.

*Proof.* The goal follows from $\mathrm{qdivideselim}(r,d)$. □

**Theorem 4.3.** *For every natural number $r$ and natural number $d$ and natural number $n$,* $\mathrm{qdivides}(r, \mathrm{wittshl}(d, n))$ *if and only if* $\mathrm{qdivides}(r, d)$.

*Proof.* The goal follows from $\mathrm{qdividesshl}(r, d, n)$. □

**Theorem 4.4.** *For every natural number $r$ and natural number $d$, if $0 < r$, then if* $\mathrm{qdivides}(r, d)$, *then* $\mathrm{vala}(d) = \mathrm{false}$ *and* $\mathrm{valb}(d) = \mathrm{false}$.

*Proof.* The goal follows from $\mathrm{qdividesval}(r, d)$. □

**Theorem 4.5.** *For every natural number $r$ and natural number $d$, if $d \neq 0$, then if* $\mathrm{qdivides}(r, d)$, *then* $2 \cdot r \leq \mathrm{binarydegree}(d)$.

*Proof.* The goal follows from $\mathrm{qdividesdegreebound}(r, d)$. □

**Theorem 4.6.** *For every natural number $d$, if $d \neq 0$, then there exists a natural number $r$ such that* $\mathrm{qdivides}(r, d)$ *and not* $\mathrm{qdivides}(r + 1, d)$.

*Proof.* The goal follows from $\mathrm{qorderexists}(d)$. □

**Theorem 4.7.** *For every natural number $r$ and natural number $d$, if $d \neq 0$, then* $\mathrm{qdivides}(r, d)$ *if and only if* $r \leq \mathrm{qorder}(d)$.

*Proof.* The goal follows from $\mathrm{qdividesifforder}(r, d)$. □

**Theorem 4.8.** *For every natural number $Q$ and natural number $t$,* $\mathrm{observersourceflux}(Q, t, 0) = 0$.

*Proof.* The goal follows from $\mathrm{observersourcefluxzero}(Q, t)$. □

**Theorem 4.9.** *For every natural number $Q$ and natural number $t$ and natural number $n$,* $\mathrm{observersourceflux}(Q, t, n{+}1) = \mathrm{wittxor}(\mathrm{wittlinear}(\mathrm{observersourceflux}(Q, t, n)), \mathrm{wittshl}(\mathrm{ob}$ $n), n + 1))$.

*Proof.* The goal follows from $\mathrm{observersourcefluxstep}(Q, t, n)$. □

**Theorem 4.10.** *For every natural number $Q$ and natural number $t$ and natural number $n$ and natural number $s$,* $\mathrm{observersourceflux}(Q, t, n + s) = \mathrm{wittxor}(\mathrm{wittlinpow}(s, \mathrm{observersourceflux}(Q, t, n)), \mathrm{wittshl}(\mathrm{observersourceflux}(Q, t{+}$ $n, s), n))$.

*Proof.* The goal follows from $\mathrm{observersourcefluxappend}(Q, t, n, s)$. □

**Theorem 4.11.** *For every natural number $Q$ and natural number $t$,* $\mathrm{observerqrank}(Q, t) = t + Q - 2 \cdot \mathrm{qorder}(\mathrm{observerdefect}(Q, t))$.

*Proof.* The goal follows from observerqrankdefinition$(Q, t)$. □

**Theorem 4.12.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if* centerperiodic$(T, P)$*, then if $T \leq t$, then* wittshl(observerdefect$(m \cdot P, t+1), 1) =$ wittxor(wittlinear(observerdefect$(m \cdot P, t)),$ wittshl(observersource$(m \cdot P, t), 1)$).

*Proof.* The goal follows from periodicobserverpolynomialstep$(T, P, m, t)$. □

**Theorem 4.13.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if* centerperiodic$(T, P)$*, then if $T \leq t$, then* observersource$(m \cdot P, t) =$ wittxor(wittxor(wittcarry(observercode$(t + m \cdot P)),$ wittcarry(observercode$(t))),$ naturalbit(leftdefect$(m \cdot P, t, 1))$).

*Proof.* The goal follows from periodicobserversource$(T, P, m, t)$. □

**Theorem 4.14.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$ and natural number $n$, if* centerperiodic$(T, P)$*, then if $T \leq t$, then* wittshl(observerdefect$(m \cdot P, t+n), n) =$ wittxor(wittlinpow$(n,$ observerdefect$(m \cdot P, t)),$ observersourceflux$(m \cdot P, t, n))$.

*Proof.* The goal follows from periodicobserverpolynomialwindow$(T, P, m, t, n)$. □

**Theorem 4.15.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$ and natural number $n$ and natural number $r$ and natural number $k$, if* centerperiodic$(T, P)$*, then if $T \leq t$, then if* qdivides$(r,$ observerdefect$(m \cdot P, t))$*, then if $k \leq n+r$, then* qdivides$(k,$ observerdefect$(m \cdot P, t + n))$ *if and only if* qdivides$(k,$ observersourceflux$(m \cdot P, t, n))$.

*Proof.* The goal follows from periodicobserverwindowdivisibility$(T, P, m, t, n, r, k)$. □

**Theorem 4.16.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$ and natural number $r$, if* centerperiodic$(T, P)$*, then if $T \leq t$, then if* qdivides$(r,$ observerdefect$(m \cdot P, t))$*, then* qdivides$(r + 1,$ observerdefect$(m \cdot P, t+1))$ *if and only if* qdivides$(r+1,$ observersource$(m \cdot P, t))$.

*Proof.* The goal follows from periodicobserverqgain$(T, P, m, t, r)$. □

**Theorem 4.17.** *For every natural number $Q$ and natural number $t$, if $0 < Q$, then $2 \cdot$* qorder(observerdefect$(Q, t)) \leq t + Q$.

*Proof.* The goal follows from observerqorderbound$(Q, t)$. □

**Theorem 4.18.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$ and natural number $n$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $T \leq t$, then if $0 < n$, then* observerqrank$(m \cdot P, t + n) <$ observerqrank$(m \cdot P, t)$ *if and only if* qdivides(qorder(observerdefect$(m \cdot P, t)) + n/2 + 1,$ observersourceflux$(m \cdot P, t, n))$.

*Proof.* The goal follows from periodicobserverqrankdrop$(T, P, m, t, n)$. □

**Theorem 4.19.** observerdefect$(5, 2) = 216$ *and* observerdefect$(5, 3) = 412 \wedge$ qdivides$(1,$ observerdefect$(5, 2)) \wedge \neg$ qdivides$(1,$ observerdefect$(5, 3)) \wedge$ center$(2) =$ false $\wedge$ center$(7) =$ false $\wedge$ center$(3) =$ true $\wedge$ center$(8) =$ true.

*Proof.* The goal follows from observerseedqloss. □

**Theorem 4.20.** *For every natural number $r$ and natural number $a$ and natural number $d$, if* wittlinpow$(r, a) = d$*, then $a \leq d$.*

*Proof.* The goal follows from qquotientbound$(r, a, d)$. □

**Theorem 4.21.** *For every natural number $d$ and natural number $r$, if* qdivides$(r + 1, d)$*, then* qordersearch$(d, r + 1) = r + 1$.

*Proof.* The goal follows from qordersearchhit$(d, r)$. □

**Theorem 4.22.** *For every natural number $d$ and natural number $r$, if not* qdivides$(r + 1, d)$*, then* qordersearch$(d, r + 1) =$ qordersearch$(d, r)$.

*Proof.* The goal follows from qordersearchmiss$(d, r)$. □

**Theorem 4.23.** *For every natural number $d$,* qorder$(d) =$ qordersearch$(d,$ binarydegree$(d))$.

*Proof.* The goal follows from qorderdefinition$(d)$. □

**Theorem 4.24.** *For every natural number $r$ and natural number $d$, if* qdivides$(r, d)$*, then there exists a natural number $a$ such that $a \leq d$ and* wittlinpow$(r, a) = d$.

*Proof.* The goal follows from qdivisesboundedwitness$(r, d)$. □

**Theorem 4.25.** *For every natural number $d$,* qordersearch$(d, 0) = 0$.

*Proof.* The goal follows from qordersearchzero$(d)$. □

**Theorem 4.26.** *For every natural number $a$ and natural number $y$,* wittlinpow(twopow($a$), $y$) = wittxor(wittxor($y$, wittshl($y$, twopow($a$))), wittshl($y$, twopow($a$)+twopow($a$))).

*Proof.* The goal follows from qfrobeniustwopow($a$, $y$). □

**Theorem 4.27.** wittlinpow(16, 1) = wittxor(wittxor(1, wittshl(1, 16)), wittshl(1, 32)) *and* wittlinpow(16, wittxor(1, wittshl(1, 16))) = wittxor(1, wittshl(1, 48)).

*Proof.* The goal follows from qsixteenpage. □

**Theorem 4.28.** *For every natural number $a$ and natural number $y$,* qdivides(twopow($a$), wittxor($y$, witts twopow($a$)))).

*Proof.* The goal follows from qdividespageshifttwopow($a$, $y$). □

**Theorem 4.29.** *For every natural number $a$ and natural number $n$ and natural number $H$,* qdivides(twopow($a$), wittxor($H$, wittxor(pagefold(3·twopow($a$), $H$, $n$), pagetail(3· twopow($a$), $H$, $n$)))).

*Proof.* The goal follows from qdividespagefoldtwopow($a$, $n$, $H$). □

**Theorem 4.30.** *For every natural number $L$ and natural number $H$ and natural number $p$, if $0 < L$, then* page($L$, wittcarry($H$), $p$) = wittxor(wittcarry(page($L$, $H$, $p$)), naturalbit(pa

*Proof.* The goal follows from pagecarry($L$, $H$, $p$). □

**Theorem 4.31.** *For every natural number $L$ and natural number $H$ and natural number $n$, if $0 < L$, then* pagefold($L$, wittcarry($H$), $n$) = carryledger($L$, $H$, $n$).

*Proof.* The goal follows from carryfoldeqledger($L$, $H$, $n$). □

**Theorem 4.32.** *For every natural number $a$ and natural number $H$ and natural number $n$,* qdivides(twopow($a$), wittxor(wittcarry($H$), wittxor(carryledger(3· twopow($a$), $H$, $n$), pagetail(3 · twopow($a$), wittcarry($H$), $n$)))).

*Proof.* The goal follows from carryledgercongruencetwopow($a$, $H$, $n$). □

**Theorem 4.33.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$ and natural number $a$, if* centerperiodic($T$, $P$), *then if $T \leq t$, then* qdivides(twopow($a$), wittxor(observersource($m$·$P$, $t$), wittxor(wittxor(carryledger(3· twopow($a$), observercode($t$+$m$·$P$), $t$+$m$·$P$+1), carryledger(3·twopow($a$), observercode($t$), $t$+ $m \cdot P + 1$)), naturalbit(leftdefect($m \cdot P$, $t$, 1))))).

*Proof.* The goal follows from periodicobserversourceledger($T$, $P$, $m$, $t$, $a$). □

**Theorem 4.34.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$ and natural number $r$ and natural number $a$, if* centerperiodic$(T, P)$*, then if* $T \leq t$*, then if* $r+1 \leq$ twopow$(a)$*, then if* qdivides$(r,$ observerdefect$(m \cdot P, t))$*, then* qdivides$(r+1,$ observerdefect$(m \cdot P, t+1))$ *if and only if* qdivides$(r+1,$ wittxor$($wittxor$($carryledger$(3 \cdot$twopow$(a),$ observercode$(t+ m \cdot P), t + m \cdot P + 1),$ carryledger$(3 \cdot$ twopow$(a),$ observercode$(t), t + m \cdot P + 1)),$ naturalbit$($leftdefect$(m \cdot P, t, 1))))$.

*Proof.* The goal follows from periodicobserverqgainledger$(T, P, m, t, r, a)$. □

**Theorem 4.35.** qdivides$(16,$ wittlinpow$(16, 1))$ *and* wittxor$($wittphi$($wittxor$(2,$ wittlinpow$(16, 1))),$ witt wittxor$($wittlinpow$(17, 1), 4) \wedge \neg$ qdivides$(16,$ wittxor$($wittphi$($wittxor$(2,$ wittlinpow$(16, 1))),$ wittphi$(2)))$

*Proof.* The goal follows from pageresiduenotclosed. □

**Theorem 4.36.** *For every natural number $a$ and natural number $A0$ and natural number $A1$ and natural number $A2$,* qdivides$($twopow$(a),$ wittxor$($wittxor$($wittxor$(A0,$ wittshl$(A1$ twopow$(a))),$ wittxor$($wittxor$(A0, A2),$ wittshl$($wittxor$(A1, A2),$ twopow$(a)))))$.

*Proof.* The goal follows from qblockresidue$(a, A0, A1, A2)$. □

**Theorem 4.37.** *For every natural number $a$ and natural number $A0$ and natural number $A1$ and natural number $A2$, if $A0 <$* twopow$($twopow$(a))$*, then if $A1 <$* twopow$($twopow$(a))$*, then if $A2 <$* twopow$($twopow$(a))$*, then* qdivides$($twopow$(a),$ wittxor$($wittxor$(A0,$ wittshl$(A1,$ twopow$(a))),$ wittshl$(A2,$ twopow$(a)+$ twopow$(a)))$ *if and only if $A0 = A2$ and $A1 = A2$.*

*Proof.* The goal follows from qblockchargezeroiff$(a, A0, A1, A2)$. □

**Theorem 4.38.** *For every natural number $a$ and natural number $A$, if $A <$* twopow$($twopow$(a))$*, then* wittcarry$($wittxor$($wittxor$(A,$ wittshl$(A,$ twopow$(a))),$ wittshl$(A,$ twopow$(a)+$ twopow$(a))))$ = wittxor$($wittxor$($wittxor$($wittcarry$(A),$ wittshl$($wittcarry$(A),$ twopow$(a))),$ wittshl$($witt twopow$(a))),$ wittxor$($wittshl$($naturalbit$($blockendpoint$($twopow$(a), A)),$ twopow$(a)),$ wittshl$($naturalbit twopow$(a)))))$.

*Proof.* The goal follows from repeatedblockcarry$(a, A)$. □

**Theorem 4.39.** *For every natural number $a$ and natural number $A$, if $A <$* twopow$($twopow$(a))$*, then* qdivides$($twopow$(a),$ wittxor$($wittcarry$($wittxor$($wittxor$(A,$ wittshl$(A,$ twopow twopow$(a)))),$ naturalbit$($blockendpoint$($twopow$(a), A))))$.

*Proof.* The goal follows from repeatedblockcarrycongruence$(a, A)$. □

**Theorem 4.40.** *For every natural number $a$ and natural number $H$ and natural number $p$ and natural number $A$, if $A <$ twopow(twopow($a$)), then if* page(3·twopow($a$), $H$, $p$) = wittxor(wittxor($A$, wittshl($A$, twopow($a$))), wittshl($A$, twopow($a$)+ twopow($a$))), *then* qdivides(twopow($a$), wittxor(page(3·twopow($a$), wittcarry($H$), $p$), wittxor(naturalbit twopow($a$), $H$, $p$))))).

*Proof.* The goal follows from pagecarryrepeated($a$, $H$, $p$, $A$). □

**Theorem 4.41.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if* centerperiodic($T$, $P$), *then if $T \leq t$, then* observersource($m \cdot P$, $t$) = wittxor(wittxor(wittcarry(observerdefect($m \cdot P$, $t$)), crosscarry(observercode($t$), observerdefect($m \cdot P$, $t$))), naturalbit(leftdefect($m \cdot P$, $t$, 1))).

*Proof.* The goal follows from periodicobserversourcecrosscarry($T$, $P$, $m$, $t$). □

**Theorem 4.42.** *There exists a natural number $s$ such that there exists a natural number $s2$ such that* qdivides(16, wittxor($s$, $s2$)) *and not* qdivides(16, wittxor(wittphi($s$), wittphi($s2$)

*Proof.* The goal follows from residuenotcongruence. □

**Theorem 4.43.** *For every natural number $Q$ and natural number $t$ and natural number $j$,* rowbit(observerdefect($Q$, $t$), $j$) = leftdefect($Q$, $t$, $j$).

*Proof.* The goal follows from observerdefectleftreadout($Q$, $t$, $j$). □

**Theorem 4.44.** *For every natural number $r$ and natural number $e$ and natural number $k$, if $e \neq 0$, then if* qdivides($r$, wittshl($e$, $k$)), *then $k + 2 \cdot r \leq$* binarydegree(wittshl($e$, $k$)).

*Proof.* The goal follows from qfrontdegreebudget($r$, $e$, $k$). □

**Theorem 4.45.** *For every natural number $n$ and natural number $e$ and natural number $o$, if $e \neq 0$, then if* twopow($n$) $\leq o$, *then if* wittshl($e$, $o$) $<$ twopow(3 · twopow($n$)), *then not* qdivides(twopow($n$), wittshl($e$, $o$)).

*Proof.* The goal follows from qpageoffsetcharge($n$, $e$, $o$). □

**Theorem 4.46.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $0 < Q$, then if* rightmostleft($Q$, $t$, $k$), *then $k + 2 \cdot$* qorder(observerdefect($Q$, $t$)) $\leq t + Q$.

*Proof.* The goal follows from observerfrontqbudget($Q$, $t$, $k$). □

**Theorem 4.47.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $0 < Q$, then if* rightmostleft$(Q, t, k)$*, then* $k \leq$ observerqrank$(Q, t)$.

*Proof.* The goal follows from observerfrontqrank$(Q, t, k)$. □

**Theorem 4.48.** *For every natural number $a$ and natural number $Q$ and natural number $t$ and natural number $p$ and natural number $o$, if* twopow$(a) \leq o$*, then if $o < 3 \cdot$* twopow$(a)$*, then if* rightmostleft$(Q, t, 3 \cdot$ twopow$(a) \cdot p + o)$*, then not* qdivides(twopow$(a)$, page$(3 \cdot$ twopow$(a)$, observerdefect$(Q, t), p))$.

*Proof.* The goal follows from observerfrontpagecharge$(a, Q, t, p, o)$. □

**Theorem 4.49.** *For every natural number $Q$ and natural number $t$, if* leftdefect$(Q, t, 0) =$ false*, then* leftdefect$(Q, t + 1, 0) =$ bxor(leftdefect$(Q, t, 1)$, centergate$(Q, t))$.

*Proof.* The goal follows from centerdefectcoupling$(Q, t)$. □

**Theorem 4.50.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if* centerperiodic$(T, P)$*, then if $T \leq t$, then* leftdefect$(m \cdot P, t, 1) =$ centergate$(m \cdot P, t)$.

*Proof.* The goal follows from periodiccentercoupling$(T, P, m, t)$. □

**Theorem 4.51.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if* centerperiodic$(T, P)$*, then if $T \leq t$, then if* center$(t) =$ true*, then* leftdefect$(m \cdot P, t, 1) =$ false.

*Proof.* The goal follows from periodicblacknoarrival$(T, P, m, t)$. □

**Theorem 4.52.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if* centerperiodic$(T, P)$*, then if $T \leq t$, then if* center$(t) =$ false*, then* leftdefect$(m \cdot P, t, 1) =$ rightdefect$(m \cdot P, t, 1)$.

*Proof.* The goal follows from periodicwhitemirror$(T, P, m, t)$. □

**Theorem 4.53.** *For every natural number $T$ and natural number $P$, if* center$(T + P) =$ center$(T)$*, then if for every natural number $t$, if $T \leq t$, then* leftdefect$(P, t, 1) =$ centergate$(P, t)$*, then* centerperiodic$(T, P)$.

*Proof.* The goal follows from centerperiodicofcoupling$(T, P)$. □

**Theorem 4.54.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if* centerperiodic$(T, P)$*, then if $T \leq t$, then* observersource$(m \cdot P, t) =$ wittxor(wittxor(wittcarry(observercode$(t + m \cdot P))$, wittcarry(observercode$(t))$), naturalbit(centergate$(m \cdot P, t))$).

*Proof.* The goal follows from periodicobserversourcecentergate$(T, P, m, t)$. □

**Theorem 4.55.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if* centerperiodic$(T, P)$*, then if* $T \leq t$*, then if* center$(t) =$ true*, then* observersource$(m \cdot P, t) =$ wittxor(wittcarry(observercode$(t + m \cdot P)$), wittcarry(observercode$(t)$)).

*Proof.* The goal follows from periodicblackobserversource$(T, P, m, t)$. □

**Theorem 4.56.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if* centerperiodic$(T, P)$*, then if* $T \leq t$*, then if* center$(t) =$ false*, then* observersource$(m \cdot P, t) =$ wittxor(wittxor(wittcarry(observercode$(t + m \cdot P)$), wittcarry(observercode$(t)$)), naturalbit(rightdefect$(m \cdot P, t, 1)$)).

*Proof.* The goal follows from periodicwhiteobserversource$(T, P, m, t)$. □

**Theorem 4.57.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $b$ and natural number $k$ and natural number $s$, if* centerperiodic$(T, P)$*, then if* $T \leq b$*, then if* rightmostleft$(m \cdot P, b, k)$*, then if* $s + 1 < k$*, then* leftdefect$(m \cdot P, b + s, 1) =$ false *and* observersource$(m \cdot P, b + s) =$ wittxor(wittcarry(observercode$(b + s + m \cdot P)$), wittcarry(observercode$(b + s)$)).

*Proof.* The goal follows from periodicfrontgapobserversource$(T, P, m, b, k, s)$. □

**Theorem 4.58.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $b$ and natural number $k$, if* centerperiodic$(T, P)$*, then if* $T \leq b$*, then if* rightmostleft$(m \cdot P, b, k)$*, then* leftdefect$(m \cdot P, b + (k - 1), 1) =$ true *and* center$(b+(k-1)) =$ false $\wedge$ rightdefect$(m \cdot P, b+(k-1), 1) =$ true $\wedge$ observersource$(m \cdot P, b+(k-1)) =$ wittxor(wittxor(wittcarry(observercode$(b + (k-1) + m \cdot P)$), wittcarry(observercode$(b + (k-1))$)), 1).

*Proof.* The goal follows from periodicfrontarrivalobserversource$(T, P, m, b, k)$. □

**Theorem 4.59.** *For every natural number $Q$ and natural number start, if* center$(start) =$ false*, then if* center$(start + Q) =$ false*, then if* center$(start + 1) =$ false*, then if* center$(start + 1 + Q) =$ false*, then if* rightcell$(start, 1) =$ false*, then if* rightcell$(start + Q, 1) =$ false*, then* leftgatephase$(Q, start + 1, 0) =$ rightphase$(Q, start, 2)$ *and* leftgatephase$(Q, start+1, 0) =$ centersign$(start, 1) \cdot$ rightphase$(Q, start, 2)$.

*Proof.* The goal follows from coupledarrivalgatephase$(Q, start)$. □

**Theorem 4.60.** *For every natural number $Q$ and natural number $t$, if $0 < Q$, then* rowbit(aligneddefect$(Q, t), 0) =$ true.

*Proof.* The goal follows from aligneddefectbitzero$(Q, t)$. □

**Theorem 4.61.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $b$ and natural number $k$, if* centerperiodic$(T, P)$*, then if $T \leq b$, then if* rightmostleft$(m \cdot P, b, k)$*, then* leftdefect$(m \cdot P, b + (k - 1), 1) =$ true *and* centergate$(m \cdot P, b + (k - 1)) =$ true $\wedge$ leftdefect$(m \cdot P, b + (k - 1) + 1, 0) =$ false.

*Proof.* The goal follows from periodicarrivalnocrossing$(T, P, m, b, k)$. □

**Theorem 4.62.** *For every natural number $Q$ and natural number $t$ and natural number $r$, if* leftadjacentdefect$(Q, t) =$ true*, then if* centercol$(t+r) =$ true*, then if* centercol$(t + r + Q) =$ true*, then* nearcollision$(Q, t, r + 1, 2 \cdot (r + 1))$.

*Proof.* The goal follows from unitdefectforcesnearcollisionkit$(Q, t, r)$. □

**Theorem 4.63.** *For every natural number $t$,* center$(t) =$ centercol$(t)$.

*Proof.* The goal follows from centereqkit$(t)$. □

**Theorem 4.64.** *For every natural number $k$,* $0 <$ twopow$(k)$.

*Proof.* The goal follows from twopowpos$(k)$. □

**Theorem 4.65.** *For every natural number $b$ and natural number $d$,* defectstep$(b, d) =$ wittxor(wittphi(wittxor$(b, d)$), wittphi$(b)$).

*Proof.* The goal follows from defectstepeq$(b, d)$. □

**Theorem 4.66.** *For every natural number $y$ and natural number $Q$,* wittphi(wittshl$(y, Q)) =$ wittshl(wittphi$(y), Q)$.

*Proof.* The goal follows from phishl$(y, Q)$. □

**Theorem 4.67.** *For every natural number $Q$ and natural number $t$,* wittxor(alignedbackground$(Q, t)$, ali wittrow$(t + Q)$.

*Proof.* The goal follows from reconstruct$(Q, t)$. □

**Theorem 4.68.** *For every natural number $T$ and natural number $P$, if $T \leq 512$, then if $0 < P$, then if $P \leq 128$, then there exists a natural number $t$ such that $T \leq t$ and $t < T + 32 \wedge \neg \mathrm{center}(t + P) = \mathrm{center}(t)$.*

*Proof.* The goal follows from horizonthirtytwo$(T, P)$. □

**Theorem 4.69.** *For every natural number $T$ and natural number $P$, if $T \leq 512$, then if $0 < P$, then if $P \leq 128$, then not* centerperiodic$(T, P)$.

*Proof.* The goal follows from nocenterperiod512128$(T, P)$. □

**Theorem 4.70.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$ and natural number $k$, if* centerperiodic$(T, P)$, *then if $T \leq t$, then if* leftdefect$(m \cdot P, t, k) = \mathrm{true}$, *then there exists a natural number $s$ such that $t \leq s$ and $s < t + k \wedge$* rightdefect$(m \cdot P, s, 1) = \mathrm{true}$.

*Proof.* The goal follows from leftdefectsourced$(T, P, m, t, k)$. □

**Theorem 4.71.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$ and natural number $k$, if* centerperiodic$(T, P)$, *then if $T \leq t$, then if* rightagrees$(m \cdot P, t, k)$, *then* leftdefect$(m \cdot P, t, k) = \mathrm{false}$.

*Proof.* The goal follows from leftconeofrightagrees$(T, P, m, t, k)$. □

**Theorem 4.72.** *For every natural number $Q$ and natural number $t$ and integer $p$, if* defect$(Q, t + 1, p) = \mathrm{false}$, *then if* defect$(Q, t, p) = \mathrm{false}$, *then if* defect$(Q, t, p + 1) = \mathrm{false}$, *then* defect$(Q, t, p - 1) = \mathrm{false}$.

*Proof.* The goal follows from defectleftofagree$(Q, t, p)$. □

**Theorem 4.73.** shortperiodsfail$(512, 128, 32) = \mathrm{true}$.

*Proof.* The goal follows from certificate51212832. □

**Theorem 4.74.** *For every natural number $T0$ and natural number $P0$ and natural number $H$ and natural number $T$ and natural number $P$, if* shortperiodsfail$(T0, P0, H) =$ true, *then if $T \leq T0$, then if $0 < P$, then if $P \leq P0$, then* centerdisagreeswithin$(T, P, H) =$ true.

*Proof.* The goal follows from checkerofcertificate$(T0, P0, H, T, P)$. □

**Theorem 4.75.** *For every natural number $T$ and natural number $P$ and natural number $H$, if* centerdisagreeswithin$(T, P, H) = \mathrm{true}$, *then there exists a natural number $t$ such that $T \leq t$ and $t < T + H \wedge \neg \mathrm{center}(t + P) = \mathrm{center}(t)$.*

*Proof.* The goal follows from disagreementofchecker$(T, P, H)$. □

**Theorem 4.76.** *For every natural number $T$ and natural number $P$ and natural number $H$, if* centerdisagreeswithin$(T, P, H) =$ true*, then not* centerperiodic$(T, P)$.

*Proof.* The goal follows from notcenterperiodicofchecker$(T, P, H)$. □

**Theorem 4.77.** *For every natural number $Q$ and natural number $t$,* rightdefect$(Q, t, 1) =$ defect$(Q, t, 1)$.

*Proof.* The goal follows from rightdefecteqdefect$(Q, t)$. □

**Theorem 4.78.** *For every natural number $t$,* center$(t) =$ not(ledgerdiagonal$(t)$).

*Proof.* The goal follows from centereqnotledgerdiagonal$(t)$. □

**Theorem 4.79.** *For every natural number $T$ and natural number $P$,* centerperiodic$(T, P)$ *if and only if* ledgerperiodic$(T, P)$.

*Proof.* The goal follows from centerperiodiciffledgerperiodic$(T, P)$. □

**Theorem 4.80.** *For every natural number $t$,* wittrow$(t) =$ wittxor(wittlinpow$(t, 1)$, wittwin$(0, t)$).

*Proof.* The goal follows from roweqlinearxorledger$(t)$. □

**Theorem 4.81.** *For every natural number $y$,* wittphi$(y) =$ wittxor(wittlinear$(y)$, wittshl(wittcarry$(y)$, 1

*Proof.* The goal follows from phieqlinearxorcarry$(y)$. □

**Theorem 4.82.** *For every natural number $a$ and natural number $b$,* wittlinear(wittxor$(a, b)) =$ wittxor(wittlinear$(a)$, wittlinear$(b)$).

*Proof.* The goal follows from philinearxor$(a, b)$. □

**Theorem 4.83.** *For every natural number $m$,* trailingparity$(2 \cdot m) =$ false.

*Proof.* The goal follows from trailingparityeven$(m)$. □

**Theorem 4.84.** *For every natural number $m$,* trailingparity$(2 \cdot m + 1) =$ not(trailingparity$(m)$).

*Proof.* The goal follows from trailingparityodd$(m)$. □

**Theorem 4.85.** *For every natural number $t$,* rowbit(philinpow$(1, t, 1), t) =$ true.

*Proof.* The goal follows from linearcenter$(t)$. □

**Theorem 4.86.** *For every natural number $t$,* $\mathrm{rowbit}(\mathrm{philinpow}(1, t, 1), t + 1) = \mathrm{trailingparity}(t)$.

*Proof.* The goal follows from $\mathrm{linearabovecenter}(t)$. □

**Theorem 4.87.** *For every natural number $t$, if $1 \leq t$, then* $\mathrm{rowbit}(\mathrm{philinpow}(1, t, 1), t - 1) = \mathrm{trailingparity}(t)$.

*Proof.* The goal follows from $\mathrm{linearbelowcenter}(t)$. □

**Theorem 4.88.** *For every natural number $s$ and natural number $y$,* $\mathrm{philin}(s, \mathrm{philin}(s, y)) = \mathrm{philin}(2 \cdot s, y)$.

*Proof.* The goal follows from $\mathrm{philinphilin}(s, y)$. □

**Theorem 4.89.** *For every natural number $m$ and natural number $y$,* $\mathrm{philinpow}(1, 2 \cdot m, y) = \mathrm{philinpow}(2, m, y)$.

*Proof.* The goal follows from $\mathrm{philinpowtwomul}(m, y)$. □

**Theorem 4.90.** *For every natural number $m$ and natural number $y$,* $\mathrm{wittlinpow}(m, y) = \mathrm{philinpow}(1, m, y)$.

*Proof.* The goal follows from $\mathrm{philinearpoweqphilinpow}(m, y)$. □

**Theorem 4.91.** *For every natural number $t$, if $1 \leq t$, then* $\mathrm{center}(t + 1) = \mathrm{rule}(\mathrm{bxor}(\mathrm{trailingparity}(t), \mathrm{ledgerbit}(t, t+1)), \mathrm{not}(\mathrm{ledgerbit}(t, t)), \mathrm{bxor}(\mathrm{trailingparity}(t), \mathrm{ledgerbit}(t, t - 1)))$.

*Proof.* The goal follows from $\mathrm{centersuccledgerpaper}(t)$. □

**Theorem 4.92.** *For every natural number $j$ and natural number $t$,* $\mathrm{rdiag}(j, t) = \mathrm{bxor}(\mathrm{rowbit}(\mathrm{philinpow}(1, t, 1), j), \mathrm{ledgerbit}(t, j))$.

*Proof.* The goal follows from $\mathrm{rdiageqlinearbxorledger}(j, t)$. □

**Theorem 4.93.** *For every natural number $t$,* $\mathrm{cell}(t, -1) = \mathrm{bxor}(\mathrm{trailingparity}(t), \mathrm{ledgerbit}(t, t + 1))$.

*Proof.* The goal follows from $\mathrm{celllefteqpaper}(t)$. □

**Theorem 4.94.** *For every natural number $t$, if $1 \leq t$, then* $\mathrm{cell}(t, 1) = \mathrm{bxor}(\mathrm{trailingparity}(t), \mathrm{ledgerbit}(t, t - 1))$.

*Proof.* The goal follows from $\mathrm{cellrighteqpaper}(t)$. □

**Theorem 4.95.** *For every natural number $T$ and natural number $P$, if $0 < P$, then if* centerperiodic$(T, P)$*, then* centerkernelfinite*.*

*Proof.* The goal follows from centerkernelfiniteofcenterperiodic$(T, P)$. □

**Theorem 4.96.** *For every natural number $T$ and natural number $P$, if* centerinfinitekernel*, then if $0 < P$, then not* centerperiodic$(T, P)$*.*

*Proof.* The goal follows from notcenterperiodicofinfinitekernel$(T, P)$. □

**Theorem 4.97.** *If* centerinfinitekernel*, then* problemone*.*

*Proof.* Assume $h$. The goal follows from problem1ofinfinitekernel$(h)$. □

**Theorem 4.98.** *For every natural number $k$ and natural number $r$ and natural number $t$,* kernelseq$(k, r, t{+}1) =$ bxor(bxor(kernelseq$(k, r, t)$, cell(twopow$(k)\cdot t + r$, negpow$(k)$)), response(twopow$(k) \cdot t + r, k)$)*.*

*Proof.* The goal follows from kernelseqstep$(k, r, t)$. □

**Theorem 4.99.** *For every natural number $t$ and natural number $k$, if $t <$* twopow$(k)$*, then* response$(t, k) =$ bxor(center$(t)$, center$(t +$ twopow$(k))$)*.*

*Proof.* The goal follows from responsecone$(t, k)$. □

**Theorem 4.100.** *For every natural number $k$ and natural number $r$ and natural number $t$,* kernelseq$(k + 1, r, t) =$ kernelseq$(k, r, 2 \cdot t)$*.*

*Proof.* The goal follows from kernelseqeven$(k, r, t)$. □

**Theorem 4.101.** *For every natural number $k$ and natural number $r$ and natural number $t$,* kernelseq$(k{+}1, r{+}$twopow$(k), t) =$ kernelseq$(k, r, 2{\cdot}t{+}1)$*.*

*Proof.* The goal follows from kernelseqodd$(k, r, t)$. □

**Theorem 4.102.** *For every natural number $k$ and natural number $r$ and natural number $t$, if* twopow$(k) \leq$ twopow$(k){\cdot}t{+}r$*, then* kernelseq$(k, r, t{+}1) =$ bxor(kernelseq$(k, r, t)$, bxor(bxor(cell(twopow$(k){\cdot}t{+}r$, negpow$(k)$)), rdiag(twopow$(k)\cdot t + r -$ twopow$(k)$, twopow$(k) \cdot t + r)$), wittresp(twopow$(k) \cdot t + r, k)$))*.*

*Proof.* The goal follows from kernelseqsteplate$(k, r, t)$. □

**Theorem 4.103.** *For every natural number $d$ and natural number $k$, if $d \leq k$, then* center$(d) =$ bxor(wittresp$(d +$ twopow$(k), k)$, response$(d +$ twopow$(k), k)$)*.*

*Proof.* The goal follows from centerrecoveredatscale($d, k$). □

**Theorem 4.104.** *If* nodyadicselfsimilarity*, then* centerinfinitekernel*.*

*Proof.* Assume $h$. The goal follows from centerinfinitekernelofnodyadicselfsimilarity($h$). □

**Theorem 4.105.** *If* nodyadicselfsimilarity*, then* problemone*.*

*Proof.* Assume $h$. The goal follows from problem1ofnodyadicselfsimilarity($h$). □

**Theorem 4.106.** *For every natural number* $k$*,* center(twopow($k + 1$)) = bxor(center(twopow($k$)), wittresp(twopow($k$), $k$))*.*

*Proof.* The goal follows from centertwopowsucc($k$). □

**Theorem 4.107.** *For every natural number* $a$ *and natural number* $b$*,* twopow($a + b$) = twopow($a$) $\cdot$ twopow($b$)*.*

*Proof.* The goal follows from twopowadd($a, b$). □

**Theorem 4.108.** *If* somedilationaperiodic*, then* centerinfinitekernel*.*

*Proof.* Assume $h$. The goal follows from centerinfinitekernelofdilationaperiodic($h$). □

**Theorem 4.109.** *If* somedilationaperiodic*, then* problemone*.*

*Proof.* Assume $h$. The goal follows from problem1ofdilationaperiodic($h$). □

**Theorem 4.110.** *If* dyadiccolumnaperiodic*, then* problemone*.*

*Proof.* Assume $h$. The goal follows from problem1ofdyadiccolumnaperiodic($h$). □

**Theorem 4.111.** *For every natural number* $b$ *and natural number* $a$ *and natural number* $c$*,* bpow($b, a + c$) = bpow($b, a$) $\cdot$ bpow($b, c$)*.*

*Proof.* The goal follows from bpowadd($b, a, c$). □

**Theorem 4.112.** *For every natural number* $b$*, if* $2 \leq b$*, then if* centerinfinitekernelin($b$)*, then* problemone*.*

*Proof.* The goal follows from problem1ofinfinitekernelin($b$). □

**Theorem 4.113.** *For every natural number $b$ and natural number $k$, if $2 \leq b$, then $k + 1 \leq \mathrm{bpow}(b, k)$.*

*Proof.* The goal follows from $\mathrm{bpowgesuccpaper}(b, k)$. □

**Theorem 4.114.** *For every natural number $b$ and natural number $T$ and natural number $P$, if $2 \leq b$, then if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$*, then* $\mathrm{centerkernelfinitein}(b)$.

*Proof.* The goal follows from $\mathrm{centerkernelfiniteinofcenterperiodicpaper}(b, T, P)$. □

**Theorem 4.115.** *For every natural number $b$ and natural number $t$, if $2 \leq b$, then if* $\mathrm{dilationaperiodicin}(b, t)$*, then* $\mathrm{centerinfinitekernelin}(b)$.

*Proof.* The goal follows from $\mathrm{centerinfinitekernelinofdilationaperiodicpaper}(b, t)$. □

**Theorem 4.116.** *For every natural number $b$ and natural number $t$, if $2 \leq b$, then if* $\mathrm{dilationaperiodicin}(b, t)$*, then* problemone.

*Proof.* The goal follows from $\mathrm{problem1ofdilationaperiodicinpaper}(b, t)$. □

**Theorem 4.117.** *For every natural number $K$,* $\mathrm{kappacollapse}(K)$ *if and only if* $\mathrm{centerperiodic}(0, \mathrm{twopow}(K))$.

*Proof.* The goal follows from $\mathrm{kappacollapseiff}(K)$. □

**Theorem 4.118.** *For every natural number $K$, if* $\mathrm{kappacollapse}(K)$*, then* $\mathrm{centerperiodic}(0, \mathrm{twopow}(K))$.

*Proof.* The goal follows from $\mathrm{centerperiodicofkappacollapse}(K)$. □

**Theorem 4.119.** *For every natural number $K$, if* $\mathrm{centerperiodic}(0, \mathrm{twopow}(K))$*, then* $\mathrm{kappacollapse}(K)$.

*Proof.* The goal follows from $\mathrm{kappacollapseofcenterperiodic}(K)$. □

**Theorem 4.120.** *For every natural number $K$, if $K \leq 7$, then not* $\mathrm{kappacollapse}(K)$.

*Proof.* The goal follows from $\mathrm{nokappacollapsesmall}(K)$. □

**Theorem 4.121.** *For every natural number $K$, if* problemone*, then not* $\mathrm{kappacollapse}(K)$.

*Proof.* The goal follows from $\mathrm{nokappacollapseofproblem1paper}(K)$. □

**Theorem 4.122.** *For every natural number $j$,* $\mathrm{center}(\mathrm{twopow}(j+1)) = \mathrm{bxor}(\mathrm{not}(\mathrm{center}(\mathrm{twopow}(j))), \mathrm{response}(\mathrm{twopow}(j), j))$.

*Proof.* The goal follows from $\mathrm{dyadiccolumnstep}(j)$. □

**Theorem 4.123.** *For every natural number $K$ and natural number $d$, if* $\mathrm{dilationperiodic}(1, K, d)$, *then* $\mathrm{dyadicresponseperiodic}(K, d)$.

*Proof.* The goal follows from $\mathrm{dyadicresponseperiodicofdilationperiodic}(K, d)$. □

**Theorem 4.124.** *If* dyadicresponseaperiodic, *then* dyadiccolumnaperiodic.

*Proof.* Assume $h$. The goal follows from $\mathrm{dyadiccolumnaperiodicofresponse}(h)$. □

**Theorem 4.125.** *If* dyadicresponseaperiodic, *then* problemone.

*Proof.* Assume $h$. The goal follows from $\mathrm{problem1ofdyadicresponseaperiodic}(h)$. □

**Theorem 4.126.** *For every natural number $j$,* $\mathrm{response}(\mathrm{twopow}(j+1), j+1) = \mathrm{bxor}(\mathrm{bxor}(\mathrm{window}(\mathrm{twopow}(j+1), \mathrm{twopow}(j), 0), \mathrm{window}(\mathrm{twopow}(j+1), \mathrm{twopow}(j), \mathrm{negpow}(j))), \mathrm{window}(\mathrm{twopow}(j+1)+\mathrm{twopow}(j), \mathrm{twopow}(j), 0))$.

*Proof.* The goal follows from $\mathrm{dyadicresponsehalvespaper}(j)$. □

**Theorem 4.127.** *For every natural number $K$ and natural number $d$, if* $\mathrm{dyadicresponseperiodic}(K, d)$, *then* $\mathrm{dilationperiodic}(1, K, 2 \cdot d)$.

*Proof.* The goal follows from $\mathrm{dilationperiodicofdyadicresponseperiodic}(K, d)$. □

**Theorem 4.128.** *If* dyadiccolumnaperiodic, *then* dyadicresponseaperiodic.

*Proof.* Assume $h$. The goal follows from $\mathrm{dyadicresponseaperiodicofcolumn}(h)$. □

**Theorem 4.129.** dyadiccolumnaperiodic *if and only if* dyadicresponseaperiodic.

*Proof.* The goal follows from dyadiccolumnaperiodiciffresponse. □

**Theorem 4.130.** *For every truth value $a$ and truth value $b$ and truth value $c$,* $\mathrm{rule}(a, b, c) = \mathrm{wolfram}(30, a, b, c)$.

*Proof.* The goal follows from $\mathrm{ruleeqwolframthirty}(a, b, c)$. □

**Theorem 4.131.** *For every truth value $a$ and truth value $b$ and truth value $c$,* $\mathrm{rule}(a, b, c) = \mathrm{bxor}(\mathrm{bxor}(\mathrm{bxor}(a, b), c), b \,\mathrm{and}\, c)$.

*Proof.* The goal follows from $\mathrm{ruleeqlinearxorcarry}(a, b, c)$. □

**Theorem 4.132.** *For every natural number $t$,* $\mathrm{wittcarry}(\mathrm{wittrow}(t)) = \mathrm{wittxor}(\mathrm{edgesource}(t), \mathrm{interior}(t$

*Proof.* The goal follows from $\mathrm{carrysplit}(t)$. □

**Theorem 4.133.** *For every natural number $t$,* $\mathrm{rowbit}(\mathrm{wittcarry}(\mathrm{wittrow}(t)), 1) = \mathrm{rdiag}(1, t)$.

*Proof.* The goal follows from $\mathrm{carryrowrightpair}(t)$. □

**Theorem 4.134.** *For every natural number $t$,* $\mathrm{wittwin}(0, t) = \mathrm{wittxor}(\mathrm{edgeledger}(t), \mathrm{interiorledger}(t))$.

*Proof.* The goal follows from $\mathrm{ledgersplit}(t)$. □

**Theorem 4.135.** *For every natural number $t$,* $\mathrm{center}(t) = \mathrm{not}(\mathrm{bxor}(\mathrm{edgediagonal}(t), \mathrm{interiordiagonal}(t)$

*Proof.* The goal follows from $\mathrm{centereqnotbxordiagonals}(t)$. □

**Theorem 4.136.** *For every natural number $n$ and natural number $d$,* $\mathrm{rowbit}(\mathrm{philinpow}(1, n, 1), d) = \mathrm{trinomialparity}(n, d)$.

*Proof.* The goal follows from $\mathrm{testbitphilinpoweqtrinomialparity}(n, d)$. □

**Theorem 4.137.** *For every natural number $t$,* $\mathrm{edgediagonal}(t) = \mathrm{edgeformula}(t, t)$.

*Proof.* The goal follows from $\mathrm{edgediagonaleqformula}(t)$. □

**Theorem 4.138.** *For every natural number $t$,* $\mathrm{center}(t) = \mathrm{not}(\mathrm{bxor}(\mathrm{edgeformula}(t, t), \mathrm{interiordiagonal}(t$

*Proof.* The goal follows from $\mathrm{centereqnotformulaxorinterior}(t)$. □

**Theorem 4.139.** *For every natural number $t$,* $\mathrm{interiordiagonal}(t) = \mathrm{interiorformula}(t, t)$.

*Proof.* The goal follows from $\mathrm{interiordiagonaleqformula}(t)$. □

**Theorem 4.140.** *For every natural number $t$,* $\mathrm{center}(t) = \mathrm{not}(\mathrm{bxor}(\mathrm{edgeformula}(t, t), \mathrm{interiorformula}(t,$

*Proof.* The goal follows from $\mathrm{centereqnotformulas}(t)$. □

**Theorem 4.141.** *For every natural number $T$ and natural number $P$,* $\mathrm{centerperiodic}(T, P)$ *if and only if* $\mathrm{interioragainstedge}(T, P)$.

*Proof.* The goal follows from $\mathrm{centerperiodiciffinterioragainstedge}(T, P)$. □

**Theorem 4.142.** *For every natural number $Q$ and natural number $z$ and natural number $t$, if* zerodiagonal$(z)$*, then if* settledby$(z+2, t)$*, then if* $z+3 \leq t$*, then* diagonalphase$(Q, z+3, t+1)$ + diagonalflux$(Q, z+3, t+1, 1)$ = diagonalphase$(Q, z+1, t)$ + diagonalflux$(Q, z+1, t, 1)$.

*Proof.* The goal follows from diagonalresetsuminvariant$(Q, z, t)$. □

**Theorem 4.143.** *For every natural number $n$ and natural number $d$, if* $2 \cdot n < d$*, then* trinomialparity$(n, d)$ = false.

*Proof.* The goal follows from trinomialparityeqfalseofgt$(n, d)$. □

**Theorem 4.144.** *For every natural number $t$,* edgediagonal$(t)$ = edgeformula$(t, t/2+1)$.

*Proof.* The goal follows from edgediagonalhalf$(t)$. □

**Theorem 4.145.** *For every natural number $T$ and natural number $P$ and natural number $t$, if* centerperiodic$(T, P)$*, then if* $T \leq t$*, then if* center$(t)$ = true*, then* bxor(ledgerbit$(t+P, t+P+1)$, ledgerbit$(t, t+1)$) = bxor(trailingparity$(t+P)$, trailingparity$(t)$).

*Proof.* The goal follows from periodicblackledgerabsorbs$(T, P, t)$. □

**Theorem 4.146.** *For every natural number $n$ and natural number $d$, if* $d \leq 2 \cdot n$*, then* trinomialparity$(n, d)$ = trinomialparity$(n, 2 \cdot n - d)$.

*Proof.* The goal follows from trinomialparitypalindrome$(n, d)$. □

**Theorem 4.147.** *For every natural number $n$,* trinomialparity$(n, n)$ = true.

*Proof.* The goal follows from trinomialparitycentral$(n)$. □

**Theorem 4.148.** *For every natural number $m$ and natural number $e$,* trinomialparity$(2 \cdot m, 2 \cdot e)$ = trinomialparity$(m, e)$.

*Proof.* The goal follows from trinomialparityeveneven$(m, e)$. □

**Theorem 4.149.** *For every natural number $m$ and natural number $e$,* trinomialparity$(2 \cdot m, 2 \cdot e + 1)$ = false.

*Proof.* The goal follows from trinomialparityevenodd$(m, e)$. □

**Theorem 4.150.** *For every natural number $m$ and natural number $e$,* trinomialparity$(2 \cdot m + 1, 2 \cdot e + 1)$ = trinomialparity$(m, e)$.

*Proof.* The goal follows from trinomialparityoddodd($m, e$). □

**Theorem 4.151.** *For every natural number $m$ and natural number $e$,* trinomialparity($2 \cdot m + 1, 2 \cdot e + 2$) = bxor(trinomialparity($m, e + 1$), trinomialparity($m, e$)).

*Proof.* The goal follows from trinomialparityoddevensucc($m, e$). □

**Theorem 4.152.** columnparity(0) = true.

*Proof.* The goal follows from columnparityzero. □

**Theorem 4.153.** *For every natural number $e$,* columnparity($2 \cdot e + 1$) = columnparity($e$).

*Proof.* The goal follows from columnparityodd($e$). □

**Theorem 4.154.** *For every natural number $e$,* columnparity($2 \cdot e + 2$) = not(columnparity($e$)).

*Proof.* The goal follows from columnparityevensucc($e$). □

**Theorem 4.155.** *For every natural number $N$,* sameparitycolumn($N$) = columnparity($N/2$).

*Proof.* The goal follows from sameparitycolumneq($N$). □

**Theorem 4.156.** *For every natural number $t$,* edgediagonal($t$) = timeparity($t$).

*Proof.* The goal follows from edgediagonaleqtimeparity($t$). □

**Theorem 4.157.** *For every natural number $t$,* center($t$) = not(bxor(timeparity($t$), interiordiagonal($t$))).

*Proof.* The goal follows from centereqnottimeparityxorinterior($t$). □

**Theorem 4.158.** *For every natural number $t$,* interiordiagonal($t$) = bxor(not(center($t$)), timeparity($t$)).

*Proof.* The goal follows from interiordiagonaleqcenter($t$). □

**Theorem 4.159.** *For every natural number $t$ and natural number $P$,* timeparity($t + 2 \cdot P$) = timeparity($t$).

*Proof.* The goal follows from timeparityaddtwomul($t, P$). □

**Theorem 4.160.** *For every natural number $T$ and natural number $P$, if* centerperiodic($T, P$)*, then* interiorperiodic($T, 2 \cdot P$).

*Proof.* The goal follows from interiorperiodicofcenterperiodic($T, P$). □

**Theorem 4.161.** *For every natural number $T$ and natural number $P$, if* interiorperiodic$(T, P)$*, then* centerperiodic$(T, 2 \cdot P)$.

*Proof.* The goal follows from centerperiodicofinteriorperiodic$(T, P)$. □

**Theorem 4.162.** problemone *if and only if* interioraperiodic.

*Proof.* The goal follows from problem1iffinterioraperiodic. □

**Theorem 4.163.** *If for every natural number $T$ and natural number $P$, if* $0 < P$*, then not* interiorperiodic$(T, P)$*, then* problemone.

*Proof.* Assume $h$. The goal follows from problem1ofinterioraperiodic$(h)$. □

**Theorem 4.164.** centerkernelfinite *if and only if* interiorkernelfinite.

*Proof.* The goal follows from centerkernelfiniteiffinterior. □

**Theorem 4.165.** centerinfinitekernel *if and only if not* interiorkernelfinite.

*Proof.* The goal follows from centerinfinitekerneliffinterior. □

**Theorem 4.166.** *If not* interiorkernelfinite*, then* problemone.

*Proof.* Assume $h$. The goal follows from problem1ofinteriorinfinitekernel$(h)$. □

**Theorem 4.167.** *For every natural number $n$,* trinomialparity$(n, 0) =$ true.

*Proof.* The goal follows from trinomialparityzeroright$(n)$. □

**Theorem 4.168.** *For every natural number $m$,* wittcarry(philinpow$(1, 2 \cdot m, 1)) = 0$.

*Proof.* The goal follows from linearselfintersectioneven$(m)$. □

**Theorem 4.169.** *For every natural number $m$ and natural number $i$,* rowbit(wittcarry(philinpow$(1, 2 \cdot m{+}1, 1)), 2 \cdot (i{+}1){+}1) =$ trinomialparity$(m, i{+}1)$ and not(trinomialparity$(m, i)$).

*Proof.* The goal follows from linearselfintersectionoddrise$(m, i)$. □

**Theorem 4.170.** *For every natural number $m$ and natural number $i$,* rowbit(wittcarry(philinpow$(1, 2 \cdot m{+}1, 1)), 2 \cdot i{+}2) =$ trinomialparity$(m, i)$ and not(trinomialparity$(m, i{+}1)$).

*Proof.* The goal follows from linearselfintersectionoddfall$(m, i)$. □

**Theorem 4.171.** *For every natural number $m$,* rowbit(wittcarry(philinpow$(1, 2 \cdot m + 1, 1)), 1) =$ true.

*Proof.* The goal follows from linearselfintersectionoddone$(m)$. □

**Theorem 4.172.** *For every natural number $m$,* interior$(m) =$ wittxor(lowsource$(m)$, interiorthree$(m)$).

*Proof.* The goal follows from interiorsplitthree$(m)$. □

**Theorem 4.173.** *For every natural number $t$,* interiorledger$(t) =$ wittxor(lowledger$(t)$, interiorthreeledg

*Proof.* The goal follows from interiorledgersplitthree$(t)$. □

**Theorem 4.174.** *For every natural number $t$,* interiordiagonal$(t) =$ bxor(lowdiagonal$(t)$, interiorthreedi

*Proof.* The goal follows from interiordiagonalsplitthree$(t)$. □

**Theorem 4.175.** *For every natural number $e$,* bxor(columnparity$(e+1)$, columnparity$(e)) =$ not(trailingparity$(e + 1)$).

*Proof.* The goal follows from columnparitysuccbxor$(e)$. □

**Theorem 4.176.** *For every natural number $s$,* lowdiagonal$(s + 2 + 1) =$ not(bxor(columnparity$(s)$, columnparity$(s/2))$).

*Proof.* The goal follows from lowdiagonalthree$(s)$. □

**Theorem 4.177.** *For every natural number $t$, if $2 \leq t$, then* bxor(timeparity$(t)$, lowdiagonal$(t)) =$ not(trailingparity$(t - 2)$).

*Proof.* The goal follows from lowdiagonaleq$(t)$. □

**Theorem 4.178.** *For every natural number $t$, if $2 \leq t$, then* center$(t) =$ bxor(trailingparity$(t - 2)$, interiorthreediagonal$(t)$).

*Proof.* The goal follows from centereqtrailingparitybxorthree$(t)$. □

**Theorem 4.179.** *For every natural number $T$ and natural number $P$, if $2 \leq T$, then* centerperiodic$(T, P)$ *if and only if* threeagainsttrailing$(T, P)$.

*Proof.* The goal follows from centerperiodiciffthreeagainsttrailing$(T, P)$. □

**Theorem 4.180.** problemone *if and only if* threeaperiodic.

*Proof.* The goal follows from problem1iffthreeaperiodic. □

**Theorem 4.181.** *If* threeaperiodic*, then* problemone.

*Proof.* Assume $h$. The goal follows from problem1ofthreeaperiodic($h$). □

**Theorem 4.182.** *For every natural number $T$ and natural number $P$, if $0 < P$, then not* trailingparityperiodic($T, P$).

*Proof.* The goal follows from trailingparitynotperiodic($T, P$). □

**Theorem 4.183.** *For every natural number $T$ and natural number $P$, if $0 < P$, then if* centerperiodic($T, P$)*, then not* threeperiodic($T, P$).

*Proof.* The goal follows from interiorthreeaperiodicofcenterperiodic($T, P$). □

**Theorem 4.184.** *For every natural number $k$ and natural number $s$, if $s + 1 <$* twopow($k$)*, then* trailingparity(twopow($k$) + $s$) = trailingparity($s$).

*Proof.* The goal follows from trailingparitytwopowadd($k, s$). □

**Theorem 4.185.** *For every natural number $j$ and natural number $m$ and natural number $d$, if $m <$* twopow($j$)*, then if $d <$* twopow($j$)*, then* trinomialparity(twopow($j$+ 1) − 1 − $m, d$) = trinomialparity(twopow($j$) − 1 − $m, d$).

*Proof.* The goal follows from trinomialparitydyadicstable($j, m, d$). □

**Theorem 4.186.** *For every natural number $j$ and natural number $m$ and natural number $d$, if $m <$* twopow($j$)*, then if $d <$* twopow($j$)*, then* trinomialparity(twopow($j$)− 1 − $m, d$) = stablekernel($m, d$).

*Proof.* The goal follows from trinomialparityeqstablekernel($j, m, d$). □

**Theorem 4.187.** *For every natural number $m$ and natural number $d$,* stablekernel($m, d$+ 2) = bxor(bxor(stablekernel($m$+1, $d$+2), stablekernel($m$+1, $d$+1)), stablekernel($m$+ 1, $d$)).

*Proof.* The goal follows from stablekernelsucc($m, d$). □

**Theorem 4.188.** *For every natural number $m$,* stablekernel($m$, 0) = stablekernel($m$+ 1, 0).

*Proof.* The goal follows from stablekernelsucczero($m$). □

**Theorem 4.189.** *For every natural number $m$,* stablekernel($m$, 1) = bxor(stablekernel($m$+ 1, 1), stablekernel($m + 1$, 0)).

*Proof.* The goal follows from stablekernelsuccone($m$). □

**Theorem 4.190.** *For every natural number $n$ and natural number $d$,* trinomialparity($n$+1, $d$+2) = bxor(bxor(trinomialparity($n$, $d$+2), trinomialparity($n$, $d$+1)), trinomialparity($n$, $d$)).

*Proof.* The goal follows from trinomialparityrowsucc($n$, $d$). □

**Theorem 4.191.** *For every natural number $j$, if* $1 \leq j$, *then* timeparity(twopow($j$)) = false.

*Proof.* The goal follows from timeparitytwopow($j$). □

**Theorem 4.192.** *For every natural number $j$ and natural number $n$ and natural number $y$ and natural number $t$, if* $n <$ twopow($j$), *then if* $t <$ twopow($j$), *then* conv(twopow($j$) − 1 − $n$, $y$, $t$) = stableconv($n$, $y$, $t$).

*Proof.* The goal follows from conveqstableconv($j$, $n$, $y$, $t$). □

**Theorem 4.193.** *For every natural number $j$, if* $1 \leq j$, *then* center(twopow($j$)) = not(stableformula($j$, twopow($j$))).

*Proof.* The goal follows from centertwopoweqstableformula($j$). □

**Theorem 4.194.** dyadiccolumnaperiodic *if and only if* stableformulaaperiodic.

*Proof.* The goal follows from dyadiccolumnaperiodiciffstable. □

**Theorem 4.195.** *If* stableformulaaperiodic, *then* problemone.

*Proof.* Assume $h$. The goal follows from problem1ofstableformulaaperiodic($h$). □

**Theorem 4.196.** *For every natural number $S$ and natural number $P$ and natural number $Q$, if* $0 < P$, *then if* $0 < Q$, *then not* trailingdiffperiodic($S$, $P$, $Q$).

*Proof.* The goal follows from trailingdiffnotperiodic($S$, $P$, $Q$). □

**Theorem 4.197.** *For every natural number $T$ and natural number $P$ and natural number $S$ and natural number $Q$, if* $0 < P$, *then if* $0 < Q$, *then if* centerperiodic($T$, $P$), *then not* threeperiodic($S$, $Q$).

*Proof.* The goal follows from notthreeperiodicofcenterperiodic($T$, $P$, $S$, $Q$). □

**Theorem 4.198.** *For every natural number $T$ and natural number $P$ and natural number $k$, if* centerperiodic($T$, $P$), *then* centerperiodic($T$, twopow($k$)·$P$).

*Proof.* The goal follows from centerperiodictwopowmul$(T, P, k)$. □

**Theorem 4.199.** *For every natural number $T$ and natural number $P$ and natural number $k$, if* centerperiodic$(T, P)$*, then* threeoffclassperiodic$(k, T + 2, P)$.

*Proof.* The goal follows from threeoffclassofcenterperiodic$(T, P, k)$. □

**Theorem 4.200.** *For every natural number $T$ and natural number $P$ and natural number $k$, if* centerperiodic$(T, P)$*, then* threeclassagainsttrailing$(k, T+1, P)$.

*Proof.* The goal follows from threeclassofcenterperiodic$(T, P, k)$. □

**Theorem 4.201.** *For every natural number $a$ and natural number $b$,* wittcarry(wittxor$(a, b)$) = wittxor(wittxor(wittcarry$(a)$, wittcarry$(b)$), cross$(a, b)$).

*Proof.* The goal follows from carryxor$(a, b)$. □

**Theorem 4.202.** *For every natural number $m$,* wittcarry(wittrow$(m)$) = wittxor(wittxor(wittcarry(wittlinpow$(m, 1)$), wittcarry(wittwin$(0, m)$))), cross(wittlinpow$(m, 1)$, wittwir

*Proof.* The goal follows from carryrowsplit$(m)$. □

**Theorem 4.203.** *For every natural number $j$ and natural number $m$, if $m <$* twopow$(j)$*, then* rowterm(twopow$(j + 1), m)$ = rowterm(twopow$(j), m)$.

*Proof.* The goal follows from rowtermtwopowsucc$(j, m)$. □

**Theorem 4.204.** *For every natural number $j$ and natural number $m$, if $m <$* twopow$(j)$*, then* rowterm(twopow$(j), m)$ = dyadicrow$(m)$.

*Proof.* The goal follows from rowtermtwopoweqdyadicrow$(j, m)$. □

**Theorem 4.205.** *For every natural number $j$, if $1 \leq j$, then* center(twopow$(j+1))$ = bxor(center(twopow$(j)$), blockparity$(j)$).

*Proof.* The goal follows from centertwopowsucceq$(j)$. □

**Theorem 4.206.** *For every natural number $j$, if $1 \leq j$, then* wittresp(twopow$(j), j)$ = blockparity$(j)$.

*Proof.* The goal follows from wittrespbooltwopoweqblockparity$(j)$. □

**Theorem 4.207.** *For every natural number $j$, if $1 \leq j$, then* response(twopow$(j), j)$ = not(blockparity$(j)$).

*Proof.* The goal follows from responsetwopoweqnotblockparity($j$). □

**Theorem 4.208.** *For every natural number $K$ and natural number $d$, if* dilationperiodic$(1, K, d)$*, then* blockparityperiodic$(K + 1, d)$*.*

*Proof.* The goal follows from blockparityperiodicofdilationperiodic($K, d$). □

**Theorem 4.209.** *If* blockparityaperiodic*, then* dyadiccolumnaperiodic*.*

*Proof.* Assume $h$. The goal follows from dyadiccolumnaperiodicofblockparityaperiodic($h$). □

**Theorem 4.210.** *If* blockparityaperiodic*, then* problemone*.*

*Proof.* Assume $h$. The goal follows from problem1ofblockparityaperiodic($h$). □

**Theorem 4.211.** *For every natural number $t$, if $2 \leq t$, then* ldiag$(2, t) =$ false*.*

*Proof.* The goal follows from ldiagtwoeq($t$). □

**Theorem 4.212.** *For every natural number $t$, if $2 \leq t$, then* ldiag$(3, t) =$ timeparity$(t)$*.*

*Proof.* The goal follows from ldiagthreeeq($t$). □

**Theorem 4.213.** *For every natural number $t$, if $2 \leq t$, then* ldiag$(4, t) =$ true*.*

*Proof.* The goal follows from ldiagfoureq($t$). □

**Theorem 4.214.** *For every natural number $t$, if $3 \leq t$, then* ldiag$(5, t) =$ timeparity$(t)$*.*

*Proof.* The goal follows from ldiagfiveeq($t$). □

**Theorem 4.215.** *For every natural number $t$, if $3 \leq t$, then* ldiag$(6, t) =$ timeparity$(t)$*.*

*Proof.* The goal follows from ldiagsixeq($t$). □

**Theorem 4.216.** *For every natural number $t$,* ldiag$(7, t) =$ false*.*

*Proof.* The goal follows from ldiagseveneq($t$). □

**Theorem 4.217.** *For every natural number $n$,* trinomialparity$(n + 2, n) =$ trailingparity$((n + 2)/2)$.

*Proof.* The goal follows from trinomialparitybelowtwo$(n)$. □

**Theorem 4.218.** *For every natural number $t$,* interiorthreeledger$(t) =$ wittxor(leftbandledger$(t)$, interio

*Proof.* The goal follows from interiorthreeledgerspliteight$(t)$. □

**Theorem 4.219.** *For every natural number $t$, if $6 \leq t$, then* leftbanddiagonal$(t) =$ bxor(not(timeparity$(t)$), bxor(trailingparity$(t-4)$, trailingparity$((t-4)/2)$)).

*Proof.* The goal follows from leftbanddiagonaleq$(t)$. □

**Theorem 4.220.** *For every natural number $t$, if $6 \leq t$, then* center$(t) =$ bxor(trailingparity$(t-2)$, bxor(bxor(not(timeparity$(t)$), bxor(trailingparity$(t-4)$, trailingparity$((t-4)/2)$)), interioreightdiagonal$(t)$)).

*Proof.* The goal follows from centereqleftband$(t)$. □

**Theorem 4.221.** *For every natural number $t$, if $6 \leq t$, then* center$(t) =$ bxor(rulerbackground$(t)$, interioreightdiagonal$(t)$).

*Proof.* The goal follows from centereqrulerbackgroundbxoreight$(t)$. □

**Theorem 4.222.** *For every natural number $T$ and natural number $P$, if $6 \leq T$, then* centerperiodic$(T, P)$ *if and only if* eightagainstrulers$(T, P)$.

*Proof.* The goal follows from centerperiodiciffeightagainstrulers$(T, P)$. □

**Theorem 4.223.** problemone *if and only if* eightaperiodic.

*Proof.* The goal follows from problem1iffeightaperiodic. □

**Theorem 4.224.** *If* eightaperiodic*, then* problemone.

*Proof.* Assume $h$. The goal follows from problem1ofeightaperiodic$(h)$. □